

SIMULATION IDEAS

The model is assumed to be

$$y_{ij} = \langle \mathcal{X}_{ij}, \mathcal{A}(t_i) \rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij} \approx \sum_{s=1}^S \sum_{r=1}^R x_{ijrs}^* \langle \mathcal{U}_r^{(s)}, \mathcal{A}^{(s)}(t_i) \rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij} = \langle \mathbf{X}_{ij}^*, \mathbf{A}(t_i) \rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij} \quad (1)$$

where matrix $\mathbf{X}_{ij}^* := (x_{ijrs}^*) \in \mathbb{R}^{R \times S}$, and $\mathbf{A}(t) := (\alpha_{rs}(t)) \in \mathbb{R}^{R \times S}$ with $\alpha_{rs}(t) := \langle \mathcal{U}_r^{(s)}, \mathcal{A}^{(s)}(t) \rangle$. Within one patient, the eyes are correlated by

$$\begin{bmatrix} \epsilon_{i1} \\ \epsilon_{i2} \end{bmatrix} \sim \mathbf{N} \left(\mathbf{0}, \sigma^2(t) \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix} \right).$$

Case 1: 2D Tensor Covariate

DGP. The tensor size is set to be $D = 2$, $p_1 = p_2 = 60$, and split by $p'_1 = p'_2 = 12$ such that by $S^{(1)} = S^{(2)} = 5$. Set the rank $R = 4$, and the s -th block tensor for j -th eye of i -th patient is generated by:

$$\mathcal{X}_{ij}^{(s)} = \sum_{r=1}^R x_{ijrs}^* \mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)}, \text{ where } x_{ijrs}^* \stackrel{\text{iid}}{\sim} \mathbf{N}(0, 1).$$

and any $\mathbf{x}_r^{(sd)}$ and $\mathbf{x}_{r'}^{(sd)}$ are orthonormal and obtained from the QR decomposition of a random matrix. The coefficient function is set to be $\mathcal{A}^{(s)}(t) = \sum_{r=1}^R \alpha_{rs}(t) \mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)}$ where $\alpha_{rs}(t)$ has 4 choices:

$$\begin{aligned} \alpha_{1rs}(t) &= 5\sqrt{\frac{rs}{RS}}(t-0.2)^2, \quad \alpha_{2rs}(t) = \sqrt{\frac{rs}{RS}} \left(e^{-(3t-1)^2} - 0.75 \right), \quad \alpha_{3rs}(t) = \sqrt{\frac{rs}{RS}} \sin(2\pi(t-0.5)), \\ \alpha_{4rs}(t) &= \sqrt{\frac{rs}{RS}} \left[0.45 \cdot 5(t-0.2)^2 + 0.35(e^{-(3t-1)^2} - 0.75) + 0.20 \sin(2\pi(t-0.5)) \right], \\ \alpha_{5rs}(t) &= \sqrt{\frac{rs}{RS}} \left[1.10 \exp \left\{ -\frac{(t-0.30)^2}{2(0.08)^2} \right\} - 0.95 \exp \left\{ -\frac{(t-0.72)^2}{2(0.11)^2} \right\} \right], \\ \alpha_{6rs}(t) &= \sqrt{\frac{rs}{RS}} \left[18(t-0.2)(t-0.55)(t-0.85) - 0.05 \right] \end{aligned}$$

The one-way vector $\mathbf{Z} \in \mathbb{R}^{n \times p_0}$ is Gaussian ensemble matrix (iid entries) and $\boldsymbol{\beta} = (2.0, 1.0, -1.0, 0.5)^\top$. We will vary $\sigma^2(t)$ to 4 subcases as Case 1(a)-(d).

$$\sigma_0^2(t) \equiv 1, \quad \sigma_1^2(t) = 1 + 0.3 \sin(2\pi t), \quad \sigma_2^2(t) = 1 + 0.25 \cos(2\pi t) + 0.1 \sin(4\pi t), \quad \text{and} \quad \sigma_3^2(t) = 0.5 + 0.5 \sin^2(\pi t). \quad (2)$$

Estimation Steps. (1) We treat the paired data as iid, and get initial estimate $\hat{\mathbf{A}}^\dagger(t)$ and $\hat{\boldsymbol{\beta}}^\dagger$. (2) Then estimate covariance structure $\hat{\sigma}^2(t)$ by eq(15)-(16) in AOAS draft. (3) Get final estimates $\hat{\mathbf{A}}^*(t)$ and $\hat{\boldsymbol{\beta}}^*$ from eq(14)-(15) of AOAS draft.

Simulation Results. We vary $n = 1000/2000$, coefficients $\alpha_{1rs}(t), \dots, \alpha_{4rs}(t)$, and correlation $\rho = 0/0.3/0.6/0.9$. We report the error of both initial estimates $(\hat{\mathbf{A}}^\dagger, \hat{\boldsymbol{\beta}}^\dagger)$ and final estimates $(\hat{\mathbf{A}}^*, \hat{\boldsymbol{\beta}}^*)$, also the error of $(\hat{\sigma}^2, \hat{\rho})$.

Table 1: MAE errors ($\times 10^{-2}$) for Case 1(a) - $\sigma_0(t) \equiv 1$.

function	n	ρ	MIAE		MAE		MIAE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
$\mathbf{A}_1(t)$	1000	0.0	5.06 ($\pm$ 0.17)	5.08 ($\pm$ 0.18)	2.09 ($\pm$ 0.78)	2.12 ($\pm$ 0.77)	30.32 ($\pm$ 2.66)	2.81 ($\pm$ 1.91)
		0.3	5.05 ($\pm$ 0.20)	4.94 ($\pm$ 0.18)	2.44 ($\pm$ 1.01)	2.49 ($\pm$ 1.07)	30.23 ($\pm$ 3.48)	7.48 ($\pm$ 3.06)
		0.6	5.06 ($\pm$ 0.24)	4.53 ($\pm$ 0.17)	2.66 ($\pm$ 1.10)	2.69 ($\pm$ 1.13)	30.19 ($\pm$ 3.71)	15.09 ($\pm$ 2.55)
		0.9	5.06 ($\pm$ 0.27)	3.57 ($\pm$ 0.18)	2.88 ($\pm$ 1.13)	2.84 ($\pm$ 1.15)	30.14 ($\pm$ 3.99)	22.70 ($\pm$ 1.92)
	2000	0.0	3.29 ($\pm$ 0.10)	3.30 ($\pm$ 0.10)	1.25 ($\pm$ 0.43)	1.24 ($\pm$ 0.44)	13.31 ($\pm$ 1.88)	1.37 ($\pm$ 0.80)
		0.3	3.32 ($\pm$ 0.11)	3.22 ($\pm$ 0.11)	1.39 ($\pm$ 0.59)	1.36 ($\pm$ 0.61)	13.24 ($\pm$ 1.82)	4.12 ($\pm$ 1.64)
		0.6	3.33 ($\pm$ 0.11)	2.89 ($\pm$ 0.10)	1.53 ($\pm$ 0.65)	1.47 ($\pm$ 0.67)	13.15 ($\pm$ 2.04)	8.48 ($\pm$ 1.41)
		0.9	3.33 ($\pm$ 0.11)	2.16 ($\pm$ 0.07)	1.68 ($\pm$ 0.70)	1.56 ($\pm$ 0.72)	13.07 ($\pm$ 2.32)	12.92 ($\pm$ 1.03)
$\mathbf{A}_2(t)$	1000	0.0	5.08 ($\pm$ 0.21)	5.10 ($\pm$ 0.21)	2.09 ($\pm$ 0.79)	2.12 ($\pm$ 0.78)	30.33 ($\pm$ 2.72)	2.94 ($\pm$ 2.11)
		0.3	5.03 ($\pm$ 0.18)	4.92 ($\pm$ 0.16)	2.44 ($\pm$ 0.98)	2.51 ($\pm$ 1.03)	30.28 ($\pm$ 3.56)	6.87 ($\pm$ 2.92)
		0.6	5.03 ($\pm$ 0.20)	4.50 ($\pm$ 0.16)	2.66 ($\pm$ 1.08)	2.71 ($\pm$ 1.11)	30.25 ($\pm$ 3.78)	14.46 ($\pm$ 2.31)
		0.9	5.02 ($\pm$ 0.22)	3.52 ($\pm$ 0.15)	2.89 ($\pm$ 1.12)	2.84 ($\pm$ 1.14)	30.20 ($\pm$ 4.05)	22.09 ($\pm$ 1.48)
	2000	0.0	3.28 ($\pm$ 0.08)	3.29 ($\pm$ 0.08)	1.21 ($\pm$ 0.46)	1.21 ($\pm$ 0.46)	13.43 ($\pm$ 1.89)	1.56 ($\pm$ 0.96)
		0.3	3.29 ($\pm$ 0.12)	3.19 ($\pm$ 0.11)	1.39 ($\pm$ 0.59)	1.37 ($\pm$ 0.63)	13.34 ($\pm$ 1.78)	3.87 ($\pm$ 1.59)
		0.6	3.29 ($\pm$ 0.12)	2.86 ($\pm$ 0.10)	1.54 ($\pm$ 0.64)	1.48 ($\pm$ 0.70)	13.24 ($\pm$ 2.03)	8.22 ($\pm$ 1.31)
		0.9	3.29 ($\pm$ 0.11)	2.09 ($\pm$ 0.08)	1.69 ($\pm$ 0.69)	1.60 ($\pm$ 0.74)	13.12 ($\pm$ 2.34)	12.64 ($\pm$ 0.98)
$\mathbf{A}_3(t)$	1000	0.0	6.49 ($\pm$ 0.22)	6.60 ($\pm$ 0.23)	2.21 ($\pm$ 0.76)	2.27 ($\pm$ 0.76)	17.16 ($\pm$ 2.64)	2.85 ($\pm$ 2.21)
		0.3	6.52 ($\pm$ 0.18)	6.56 ($\pm$ 0.17)	2.46 ($\pm$ 1.11)	2.53 ($\pm$ 1.09)	17.08 ($\pm$ 3.21)	11.00 ($\pm$ 3.08)
		0.6	6.53 ($\pm$ 0.19)	6.29 ($\pm$ 0.16)	2.70 ($\pm$ 1.18)	2.73 ($\pm$ 1.16)	17.15 ($\pm$ 3.26)	22.32 ($\pm$ 2.77)
		0.9	6.54 ($\pm$ 0.22)	5.75 ($\pm$ 0.15)	2.95 ($\pm$ 1.21)	2.89 ($\pm$ 1.22)	17.24 ($\pm$ 3.35)	33.66 ($\pm$ 2.46)
	2000	0.0	5.04 ($\pm$ 0.11)	5.14 ($\pm$ 0.12)	1.49 ($\pm$ 0.56)	1.48 ($\pm$ 0.58)	9.82 ($\pm$ 1.17)	1.33 ($\pm$ 0.98)
		0.3	5.05 ($\pm$ 0.12)	5.09 ($\pm$ 0.12)	1.41 ($\pm$ 0.59)	1.38 ($\pm$ 0.61)	10.29 ($\pm$ 1.30)	8.32 ($\pm$ 1.90)
		0.6	5.06 ($\pm$ 0.13)	4.89 ($\pm$ 0.11)	1.55 ($\pm$ 0.65)	1.49 ($\pm$ 0.66)	10.41 ($\pm$ 1.36)	16.81 ($\pm$ 1.83)
		0.9	5.06 ($\pm$ 0.13)	4.50 ($\pm$ 0.09)	1.67 ($\pm$ 0.70)	1.58 ($\pm$ 0.70)	10.57 ($\pm$ 1.48)	25.32 ($\pm$ 1.67)
$\mathbf{A}_4(t)$	1000	0.0	4.87 ($\pm$ 0.19)	4.88 ($\pm$ 0.20)	2.09 ($\pm$ 0.76)	2.13 ($\pm$ 0.75)	31.77 ($\pm$ 2.69)	2.85 ($\pm$ 1.97)
		0.3	4.85 ($\pm$ 0.18)	4.73 ($\pm$ 0.16)	2.43 ($\pm$ 1.01)	2.47 ($\pm$ 1.07)	31.68 ($\pm$ 3.47)	6.83 ($\pm$ 2.97)
		0.6	4.85 ($\pm$ 0.21)	4.28 ($\pm$ 0.15)	2.65 ($\pm$ 1.10)	2.67 ($\pm$ 1.14)	31.64 ($\pm$ 3.69)	13.98 ($\pm$ 2.37)
		0.9	4.86 ($\pm$ 0.24)	3.20 ($\pm$ 0.16)	2.88 ($\pm$ 1.13)	2.82 ($\pm$ 1.15)	31.59 ($\pm$ 3.97)	21.14 ($\pm$ 1.64)
	2000	0.0	3.02 ($\pm$ 0.09)	3.02 ($\pm$ 0.09)	1.26 ($\pm$ 0.45)	1.26 ($\pm$ 0.45)	15.12 ($\pm$ 1.87)	1.44 ($\pm$ 0.78)
		0.3	3.03 ($\pm$ 0.09)	2.91 ($\pm$ 0.09)	1.36 ($\pm$ 0.58)	1.33 ($\pm$ 0.60)	15.04 ($\pm$ 1.85)	3.57 ($\pm$ 1.45)
		0.6	3.04 ($\pm$ 0.09)	2.54 ($\pm$ 0.08)	1.51 ($\pm$ 0.64)	1.45 ($\pm$ 0.66)	14.95 ($\pm$ 2.06)	7.33 ($\pm$ 1.30)
		0.9	3.04 ($\pm$ 0.09)	1.63 ($\pm$ 0.06)	1.65 ($\pm$ 0.69)	1.54 ($\pm$ 0.71)	14.85 ($\pm$ 2.35)	11.21 ($\pm$ 0.88)
$\mathbf{A}_5(t)$	1000	0.0	7.88 ($\pm$ 0.22)	8.19 ($\pm$ 0.24)	2.56 ($\pm$ 1.05)	2.49 ($\pm$ 1.00)	28.32 ($\pm$ 2.37)	3.50 ($\pm$ 2.04)
		0.3	7.84 ($\pm$ 0.26)	8.09 ($\pm$ 0.26)	2.63 ($\pm$ 0.83)	2.71 ($\pm$ 0.96)	28.28 ($\pm$ 2.54)	14.55 ($\pm$ 3.56)
		0.6	7.85 ($\pm$ 0.27)	7.89 ($\pm$ 0.26)	2.85 ($\pm$ 0.88)	2.93 ($\pm$ 1.03)	28.21 ($\pm$ 2.58)	29.92 ($\pm$ 3.58)
		0.9	7.85 ($\pm$ 0.30)	7.48 ($\pm$ 0.28)	3.08 ($\pm$ 0.93)	3.13 ($\pm$ 1.11)	28.19 ($\pm$ 2.66)	45.30 ($\pm$ 3.57)
	2000	0.0	6.46 ($\pm$ 0.14)	6.77 ($\pm$ 0.14)	1.59 ($\pm$ 0.55)	1.53 ($\pm$ 0.50)	38.31 ($\pm$ 2.77)	2.00 ($\pm$ 1.43)
		0.3	6.48 ($\pm$ 0.13)	6.73 ($\pm$ 0.13)	1.66 ($\pm$ 0.63)	1.61 ($\pm$ 0.55)	37.98 ($\pm$ 2.08)	13.04 ($\pm$ 2.74)
		0.6	6.49 ($\pm$ 0.13)	6.57 ($\pm$ 0.13)	1.79 ($\pm$ 0.67)	1.74 ($\pm$ 0.59)	38.02 ($\pm$ 2.04)	26.38 ($\pm$ 2.59)
		0.9	6.48 ($\pm$ 0.13)	6.24 ($\pm$ 0.13)	1.93 ($\pm$ 0.71)	1.85 ($\pm$ 0.61)	38.11 ($\pm$ 2.09)	39.79 ($\pm$ 2.32)
$\mathbf{A}_6(t)$	1000	0.0	6.18 ($\pm$ 0.21)	6.26 ($\pm$ 0.21)	2.10 ($\pm$ 0.84)	2.15 ($\pm$ 0.81)	20.74 ($\pm$ 2.87)	2.91 ($\pm$ 2.08)
		0.3	6.13 ($\pm$ 0.21)	6.12 ($\pm$ 0.20)	2.62 ($\pm$ 1.01)	2.64 ($\pm$ 1.05)	20.63 ($\pm$ 3.74)	9.38 ($\pm$ 3.13)
		0.6	6.12 ($\pm$ 0.23)	5.81 ($\pm$ 0.20)	2.83 ($\pm$ 1.12)	2.82 ($\pm$ 1.13)	20.62 ($\pm$ 3.94)	19.66 ($\pm$ 2.69)
		0.9	6.11 ($\pm$ 0.25)	5.18 ($\pm$ 0.20)	3.04 ($\pm$ 1.19)	2.96 ($\pm$ 1.18)	20.62 ($\pm$ 4.18)	29.96 ($\pm$ 2.07)
	2000	0.0	4.59 ($\pm$ 0.10)	4.65 ($\pm$ 0.10)	1.17 ($\pm$ 0.45)	1.17 ($\pm$ 0.45)	6.90 ($\pm$ 1.09)	1.67 ($\pm$ 1.00)
		0.3	4.61 ($\pm$ 0.13)	4.61 ($\pm$ 0.12)	1.55 ($\pm$ 0.64)	1.54 ($\pm$ 0.67)	7.11 ($\pm$ 1.22)	6.86 ($\pm$ 1.85)
		0.6	4.61 ($\pm$ 0.13)	4.40 ($\pm$ 0.11)	1.71 ($\pm$ 0.69)	1.66 ($\pm$ 0.74)	7.16 ($\pm$ 1.31)	14.36 ($\pm$ 1.69)
		0.9	4.60 ($\pm$ 0.14)	3.95 ($\pm$ 0.08)	1.86 ($\pm$ 0.73)	1.79 ($\pm$ 0.78)	7.26 ($\pm$ 1.44)	21.94 ($\pm$ 1.52)

Table 2: MAE errors ($\times 10^{-2}$) for Case 1(b) - $\sigma_1(t) = 1 + 0.3 \sin(2\pi t)$.

function	n	ρ	MIAE		MAE		MIAE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
$\mathbf{A}_1(t)$	1000	0.0	5.05 ($\pm$ 0.18)	5.05 ($\pm$ 0.18)	2.12 ($\pm$ 0.80)	2.05 ($\pm$ 0.75)	30.43 ($\pm$ 2.75)	2.93 ($\pm$ 1.94)
		0.3	5.03 ($\pm$ 0.20)	4.92 ($\pm$ 0.18)	2.41 ($\pm$ 1.03)	2.45 ($\pm$ 1.04)	30.27 ($\pm$ 3.47)	7.63 ($\pm$ 3.19)
		0.6	5.03 ($\pm$ 0.23)	4.51 ($\pm$ 0.17)	2.63 ($\pm$ 1.10)	2.65 ($\pm$ 1.10)	30.26 ($\pm$ 3.76)	15.23 ($\pm$ 2.67)
		0.9	5.04 ($\pm$ 0.27)	3.56 ($\pm$ 0.18)	2.86 ($\pm$ 1.14)	2.79 ($\pm$ 1.12)	30.23 ($\pm$ 4.12)	22.80 ($\pm$ 2.02)
	2000	0.0	3.29 ($\pm$ 0.10)	3.29 ($\pm$ 0.10)	1.24 ($\pm$ 0.46)	1.21 ($\pm$ 0.42)	13.47 ($\pm$ 1.92)	1.60 ($\pm$ 0.90)
		0.3	3.31 ($\pm$ 0.11)	3.20 ($\pm$ 0.11)	1.39 ($\pm$ 0.61)	1.34 ($\pm$ 0.58)	13.41 ($\pm$ 1.93)	4.31 ($\pm$ 1.55)
		0.6	3.32 ($\pm$ 0.11)	2.88 ($\pm$ 0.10)	1.53 ($\pm$ 0.67)	1.44 ($\pm$ 0.65)	13.46 ($\pm$ 2.10)	8.63 ($\pm$ 1.31)
		0.9	3.32 ($\pm$ 0.11)	2.15 ($\pm$ 0.07)	1.67 ($\pm$ 0.72)	1.53 ($\pm$ 0.69)	13.57 ($\pm$ 2.25)	12.97 ($\pm$ 0.98)
$\mathbf{A}_2(t)$	1000	0.0	5.05 ($\pm$ 0.21)	5.07 ($\pm$ 0.21)	2.11 ($\pm$ 0.82)	2.06 ($\pm$ 0.75)	30.43 ($\pm$ 2.82)	3.14 ($\pm$ 2.03)
		0.3	4.99 ($\pm$ 0.18)	4.89 ($\pm$ 0.17)	2.42 ($\pm$ 1.01)	2.44 ($\pm$ 1.02)	30.32 ($\pm$ 3.54)	6.97 ($\pm$ 3.07)
		0.6	4.99 ($\pm$ 0.20)	4.47 ($\pm$ 0.16)	2.66 ($\pm$ 1.09)	2.63 ($\pm$ 1.11)	30.32 ($\pm$ 3.83)	14.55 ($\pm$ 2.42)
		0.9	4.99 ($\pm$ 0.22)	3.49 ($\pm$ 0.15)	2.88 ($\pm$ 1.15)	2.78 ($\pm$ 1.12)	30.30 ($\pm$ 4.18)	22.12 ($\pm$ 1.51)
	2000	0.0	3.26 ($\pm$ 0.08)	3.27 ($\pm$ 0.08)	1.19 ($\pm$ 0.46)	1.18 ($\pm$ 0.44)	13.55 ($\pm$ 1.99)	1.77 ($\pm$ 1.12)
		0.3	3.26 ($\pm$ 0.12)	3.17 ($\pm$ 0.12)	1.38 ($\pm$ 0.62)	1.35 ($\pm$ 0.58)	13.41 ($\pm$ 1.94)	4.06 ($\pm$ 1.53)
		0.6	3.26 ($\pm$ 0.12)	2.84 ($\pm$ 0.10)	1.54 ($\pm$ 0.67)	1.46 ($\pm$ 0.65)	13.41 ($\pm$ 2.12)	8.36 ($\pm$ 1.28)
		0.9	3.26 ($\pm$ 0.11)	2.07 ($\pm$ 0.08)	1.69 ($\pm$ 0.71)	1.56 ($\pm$ 0.70)	13.45 ($\pm$ 2.32)	12.69 ($\pm$ 1.00)
$\mathbf{A}_3(t)$	1000	0.0	6.48 ($\pm$ 0.22)	6.59 ($\pm$ 0.23)	2.22 ($\pm$ 0.79)	2.21 ($\pm$ 0.75)	19.24 ($\pm$ 2.34)	2.95 ($\pm$ 2.18)
		0.3	6.51 ($\pm$ 0.18)	6.54 ($\pm$ 0.17)	2.43 ($\pm$ 1.08)	2.49 ($\pm$ 1.08)	19.03 ($\pm$ 2.81)	11.20 ($\pm$ 3.26)
		0.6	6.52 ($\pm$ 0.20)	6.28 ($\pm$ 0.17)	2.67 ($\pm$ 1.15)	2.69 ($\pm$ 1.15)	19.03 ($\pm$ 3.09)	22.52 ($\pm$ 2.99)
		0.9	6.53 ($\pm$ 0.22)	5.74 ($\pm$ 0.16)	2.91 ($\pm$ 1.19)	2.86 ($\pm$ 1.20)	19.06 ($\pm$ 3.45)	33.85 ($\pm$ 2.68)
	2000	0.0	5.04 ($\pm$ 0.11)	5.14 ($\pm$ 0.12)	1.52 ($\pm$ 0.55)	1.47 ($\pm$ 0.56)	10.30 ($\pm$ 1.13)	1.39 ($\pm$ 1.04)
		0.3	5.05 ($\pm$ 0.12)	5.08 ($\pm$ 0.12)	1.41 ($\pm$ 0.58)	1.35 ($\pm$ 0.61)	10.84 ($\pm$ 1.29)	8.51 ($\pm$ 1.89)
		0.6	5.06 ($\pm$ 0.13)	4.89 ($\pm$ 0.11)	1.54 ($\pm$ 0.63)	1.46 ($\pm$ 0.66)	10.99 ($\pm$ 1.33)	16.96 ($\pm$ 1.81)
		0.9	5.06 ($\pm$ 0.13)	4.49 ($\pm$ 0.09)	1.67 ($\pm$ 0.68)	1.56 ($\pm$ 0.70)	11.22 ($\pm$ 1.37)	25.42 ($\pm$ 1.66)
$\mathbf{A}_4(t)$	1000	0.0	4.85 ($\pm$ 0.20)	4.86 ($\pm$ 0.20)	2.12 ($\pm$ 0.78)	2.05 ($\pm$ 0.71)	31.88 ($\pm$ 2.79)	2.95 ($\pm$ 2.01)
		0.3	4.82 ($\pm$ 0.18)	4.70 ($\pm$ 0.16)	2.39 ($\pm$ 1.03)	2.42 ($\pm$ 1.05)	31.72 ($\pm$ 3.45)	6.98 ($\pm$ 3.13)
		0.6	4.82 ($\pm$ 0.21)	4.26 ($\pm$ 0.15)	2.62 ($\pm$ 1.10)	2.61 ($\pm$ 1.12)	31.71 ($\pm$ 3.75)	14.11 ($\pm$ 2.52)
		0.9	4.82 ($\pm$ 0.24)	3.18 ($\pm$ 0.15)	2.85 ($\pm$ 1.14)	2.77 ($\pm$ 1.13)	31.69 ($\pm$ 4.10)	21.23 ($\pm$ 1.76)
	2000	0.0	3.00 ($\pm$ 0.09)	3.00 ($\pm$ 0.09)	1.25 ($\pm$ 0.47)	1.23 ($\pm$ 0.43)	15.24 ($\pm$ 1.94)	1.65 ($\pm$ 0.92)
		0.3	3.01 ($\pm$ 0.09)	2.89 ($\pm$ 0.09)	1.36 ($\pm$ 0.59)	1.31 ($\pm$ 0.57)	15.15 ($\pm$ 1.94)	3.71 ($\pm$ 1.47)
		0.6	3.02 ($\pm$ 0.09)	2.52 ($\pm$ 0.08)	1.51 ($\pm$ 0.65)	1.42 ($\pm$ 0.64)	15.14 ($\pm$ 2.11)	7.47 ($\pm$ 1.21)
		0.9	3.02 ($\pm$ 0.09)	1.62 ($\pm$ 0.06)	1.65 ($\pm$ 0.70)	1.51 ($\pm$ 0.68)	15.15 ($\pm$ 2.33)	11.25 ($\pm$ 0.85)
$\mathbf{A}_5(t)$	1000	0.0	7.85 ($\pm$ 0.23)	8.15 ($\pm$ 0.25)	2.57 ($\pm$ 1.07)	2.44 ($\pm$ 0.94)	26.70 ($\pm$ 2.27)	3.61 ($\pm$ 2.24)
		0.3	7.81 ($\pm$ 0.25)	8.05 ($\pm$ 0.25)	2.63 ($\pm$ 0.82)	2.60 ($\pm$ 0.94)	26.69 ($\pm$ 2.51)	14.59 ($\pm$ 3.55)
		0.6	7.81 ($\pm$ 0.26)	7.85 ($\pm$ 0.25)	2.85 ($\pm$ 0.86)	2.82 ($\pm$ 1.02)	26.63 ($\pm$ 2.58)	29.96 ($\pm$ 3.56)
		0.9	7.81 ($\pm$ 0.28)	7.45 ($\pm$ 0.27)	3.08 ($\pm$ 0.90)	3.02 ($\pm$ 1.08)	26.58 ($\pm$ 2.69)	45.32 ($\pm$ 3.51)
	2000	0.0	6.44 ($\pm$ 0.13)	6.74 ($\pm$ 0.14)	1.58 ($\pm$ 0.53)	1.48 ($\pm$ 0.47)	38.26 ($\pm$ 2.98)	2.13 ($\pm$ 1.50)
		0.3	6.46 ($\pm$ 0.13)	6.70 ($\pm$ 0.13)	1.67 ($\pm$ 0.63)	1.57 ($\pm$ 0.56)	37.88 ($\pm$ 2.37)	13.22 ($\pm$ 2.83)
		0.6	6.46 ($\pm$ 0.13)	6.54 ($\pm$ 0.13)	1.80 ($\pm$ 0.67)	1.69 ($\pm$ 0.60)	37.88 ($\pm$ 2.32)	26.55 ($\pm$ 2.68)
		0.9	6.46 ($\pm$ 0.13)	6.22 ($\pm$ 0.13)	1.93 ($\pm$ 0.71)	1.80 ($\pm$ 0.62)	37.90 ($\pm$ 2.35)	39.91 ($\pm$ 2.40)
$\mathbf{A}_6(t)$	1000	0.0	6.16 ($\pm$ 0.21)	6.23 ($\pm$ 0.22)	2.11 ($\pm$ 0.87)	2.11 ($\pm$ 0.79)	20.88 ($\pm$ 2.96)	3.10 ($\pm$ 2.08)
		0.3	6.10 ($\pm$ 0.21)	6.09 ($\pm$ 0.20)	2.62 ($\pm$ 1.05)	2.57 ($\pm$ 1.01)	20.76 ($\pm$ 3.59)	9.42 ($\pm$ 3.30)
		0.6	6.09 ($\pm$ 0.23)	5.78 ($\pm$ 0.20)	2.83 ($\pm$ 1.15)	2.75 ($\pm$ 1.09)	20.79 ($\pm$ 3.83)	19.70 ($\pm$ 2.82)
		0.9	6.08 ($\pm$ 0.24)	5.16 ($\pm$ 0.20)	3.04 ($\pm$ 1.23)	2.88 ($\pm$ 1.15)	20.80 ($\pm$ 4.15)	29.97 ($\pm$ 2.13)
	2000	0.0	4.58 ($\pm$ 0.10)	4.64 ($\pm$ 0.10)	1.16 ($\pm$ 0.45)	1.16 ($\pm$ 0.44)	6.98 ($\pm$ 1.29)	1.80 ($\pm$ 1.19)
		0.3	4.58 ($\pm$ 0.13)	4.59 ($\pm$ 0.12)	1.53 ($\pm$ 0.69)	1.52 ($\pm$ 0.64)	7.08 ($\pm$ 1.29)	7.02 ($\pm$ 1.88)
		0.6	4.59 ($\pm$ 0.13)	4.38 ($\pm$ 0.11)	1.68 ($\pm$ 0.75)	1.64 ($\pm$ 0.70)	7.21 ($\pm$ 1.49)	14.49 ($\pm$ 1.74)
		0.9	4.58 ($\pm$ 0.14)	3.94 ($\pm$ 0.08)	1.84 ($\pm$ 0.79)	1.75 ($\pm$ 0.74)	7.38 ($\pm$ 1.73)	22.00 ($\pm$ 1.61)

Table 3: MAE errors ($\times 10^{-2}$) for Case 1(c) - $\sigma_2(t) = 1 + 0.25 \cos(2\pi t) + 0.1 \sin(4\pi t)$.

function	n	ρ	MIAE		MAE		MIAE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
$\mathbf{A}_1(t)$	1000	0.0	4.28 ($\pm$ 0.14)	4.29 ($\pm$ 0.14)	1.82 ($\pm$ 0.68)	1.78 ($\pm$ 0.66)	20.98 ($\pm$ 2.10)	2.79 ($\pm$ 1.96)
		0.3	4.28 ($\pm$ 0.18)	4.19 ($\pm$ 0.16)	2.08 ($\pm$ 0.85)	2.12 ($\pm$ 0.89)	20.91 ($\pm$ 2.62)	7.39 ($\pm$ 3.17)
		0.6	4.28 ($\pm$ 0.20)	3.84 ($\pm$ 0.15)	2.27 ($\pm$ 0.92)	2.29 ($\pm$ 0.94)	20.89 ($\pm$ 2.80)	14.77 ($\pm$ 2.63)
		0.9	4.28 ($\pm$ 0.23)	3.06 ($\pm$ 0.15)	2.45 ($\pm$ 0.96)	2.40 ($\pm$ 0.96)	20.85 ($\pm$ 3.03)	22.15 ($\pm$ 1.96)
	2000	0.0	2.85 ($\pm$ 0.09)	2.86 ($\pm$ 0.09)	1.09 ($\pm$ 0.38)	1.04 ($\pm$ 0.38)	8.56 ($\pm$ 1.44)	1.51 ($\pm$ 0.91)
		0.3	2.87 ($\pm$ 0.09)	2.79 ($\pm$ 0.09)	1.23 ($\pm$ 0.52)	1.13 ($\pm$ 0.51)	8.47 ($\pm$ 1.43)	4.17 ($\pm$ 1.73)
		0.6	2.88 ($\pm$ 0.09)	2.53 ($\pm$ 0.09)	1.35 ($\pm$ 0.57)	1.23 ($\pm$ 0.56)	8.41 ($\pm$ 1.58)	8.51 ($\pm$ 1.42)
		0.9	2.88 ($\pm$ 0.10)	1.96 ($\pm$ 0.06)	1.48 ($\pm$ 0.61)	1.31 ($\pm$ 0.60)	8.37 ($\pm$ 1.77)	12.91 ($\pm$ 1.01)
$\mathbf{A}_2(t)$	1000	0.0	4.29 ($\pm$ 0.17)	4.31 ($\pm$ 0.17)	1.83 ($\pm$ 0.69)	1.77 ($\pm$ 0.67)	21.00 ($\pm$ 2.15)	3.03 ($\pm$ 2.04)
		0.3	4.24 ($\pm$ 0.15)	4.16 ($\pm$ 0.14)	2.08 ($\pm$ 0.83)	2.13 ($\pm$ 0.87)	20.97 ($\pm$ 2.70)	6.68 ($\pm$ 2.98)
		0.6	4.24 ($\pm$ 0.17)	3.81 ($\pm$ 0.13)	2.27 ($\pm$ 0.89)	2.30 ($\pm$ 0.93)	20.95 ($\pm$ 2.87)	14.04 ($\pm$ 2.36)
		0.9	4.23 ($\pm$ 0.18)	3.00 ($\pm$ 0.13)	2.48 ($\pm$ 0.92)	2.41 ($\pm$ 0.97)	20.91 ($\pm$ 3.09)	21.43 ($\pm$ 1.49)
	2000	0.0	2.83 ($\pm$ 0.07)	2.84 ($\pm$ 0.07)	1.04 ($\pm$ 0.40)	1.01 ($\pm$ 0.39)	8.68 ($\pm$ 1.45)	1.75 ($\pm$ 1.15)
		0.3	2.83 ($\pm$ 0.10)	2.76 ($\pm$ 0.10)	1.22 ($\pm$ 0.53)	1.15 ($\pm$ 0.51)	8.56 ($\pm$ 1.41)	3.87 ($\pm$ 1.72)
		0.6	2.83 ($\pm$ 0.10)	2.49 ($\pm$ 0.08)	1.36 ($\pm$ 0.56)	1.25 ($\pm$ 0.58)	8.49 ($\pm$ 1.58)	8.19 ($\pm$ 1.38)
		0.9	2.83 ($\pm$ 0.10)	1.87 ($\pm$ 0.06)	1.49 ($\pm$ 0.59)	1.34 ($\pm$ 0.61)	8.42 ($\pm$ 1.81)	12.56 ($\pm$ 0.99)
$\mathbf{A}_3(t)$	1000	0.0	5.84 ($\pm$ 0.19)	5.96 ($\pm$ 0.20)	1.94 ($\pm$ 0.68)	1.95 ($\pm$ 0.68)	9.43 ($\pm$ 1.33)	3.00 ($\pm$ 2.14)
		0.3	5.88 ($\pm$ 0.15)	5.94 ($\pm$ 0.15)	2.12 ($\pm$ 0.98)	2.15 ($\pm$ 0.90)	9.57 ($\pm$ 1.61)	11.74 ($\pm$ 3.09)
		0.6	5.88 ($\pm$ 0.16)	5.74 ($\pm$ 0.14)	2.32 ($\pm$ 1.05)	2.32 ($\pm$ 0.95)	9.61 ($\pm$ 1.60)	23.73 ($\pm$ 2.84)
		0.9	5.89 ($\pm$ 0.17)	5.34 ($\pm$ 0.13)	2.54 ($\pm$ 1.06)	2.45 ($\pm$ 1.00)	9.69 ($\pm$ 1.68)	35.74 ($\pm$ 2.62)
	2000	0.0	4.72 ($\pm$ 0.10)	4.84 ($\pm$ 0.11)	1.36 ($\pm$ 0.51)	1.29 ($\pm$ 0.51)	10.46 ($\pm$ 0.95)	1.38 ($\pm$ 1.05)
		0.3	4.72 ($\pm$ 0.11)	4.80 ($\pm$ 0.10)	1.28 ($\pm$ 0.50)	1.17 ($\pm$ 0.52)	10.69 ($\pm$ 1.10)	9.40 ($\pm$ 2.06)
		0.6	4.73 ($\pm$ 0.12)	4.65 ($\pm$ 0.10)	1.39 ($\pm$ 0.55)	1.26 ($\pm$ 0.56)	10.78 ($\pm$ 1.15)	18.86 ($\pm$ 1.97)
		0.9	4.73 ($\pm$ 0.12)	4.37 ($\pm$ 0.08)	1.50 ($\pm$ 0.59)	1.34 ($\pm$ 0.60)	10.92 ($\pm$ 1.25)	28.35 ($\pm$ 1.83)
$\mathbf{A}_4(t)$	1000	0.0	4.07 ($\pm$ 0.15)	4.07 ($\pm$ 0.16)	1.81 ($\pm$ 0.67)	1.79 ($\pm$ 0.64)	22.43 ($\pm$ 2.13)	2.89 ($\pm$ 1.95)
		0.3	4.05 ($\pm$ 0.15)	3.95 ($\pm$ 0.14)	2.07 ($\pm$ 0.86)	2.09 ($\pm$ 0.89)	22.37 ($\pm$ 2.61)	6.57 ($\pm$ 3.04)
		0.6	4.06 ($\pm$ 0.18)	3.57 ($\pm$ 0.13)	2.26 ($\pm$ 0.93)	2.27 ($\pm$ 0.94)	22.34 ($\pm$ 2.78)	13.34 ($\pm$ 2.41)
		0.9	4.05 ($\pm$ 0.20)	2.65 ($\pm$ 0.13)	2.46 ($\pm$ 0.96)	2.38 ($\pm$ 0.96)	22.31 ($\pm$ 3.01)	20.13 ($\pm$ 1.62)
	2000	0.0	2.55 ($\pm$ 0.08)	2.55 ($\pm$ 0.08)	1.10 ($\pm$ 0.40)	1.05 ($\pm$ 0.38)	10.37 ($\pm$ 1.44)	1.56 ($\pm$ 0.90)
		0.3	2.56 ($\pm$ 0.08)	2.45 ($\pm$ 0.08)	1.19 ($\pm$ 0.50)	1.11 ($\pm$ 0.50)	10.27 ($\pm$ 1.46)	3.45 ($\pm$ 1.54)
		0.6	2.56 ($\pm$ 0.08)	2.15 ($\pm$ 0.07)	1.33 ($\pm$ 0.55)	1.21 ($\pm$ 0.55)	10.21 ($\pm$ 1.61)	7.01 ($\pm$ 1.30)
		0.9	2.56 ($\pm$ 0.08)	1.39 ($\pm$ 0.05)	1.45 ($\pm$ 0.59)	1.28 ($\pm$ 0.59)	10.14 ($\pm$ 1.82)	10.67 ($\pm$ 0.83)
$\mathbf{A}_5(t)$	1000	0.0	7.99 ($\pm$ 0.23)	8.31 ($\pm$ 0.25)	2.56 ($\pm$ 1.06)	2.50 ($\pm$ 1.00)	32.41 ($\pm$ 2.33)	3.50 ($\pm$ 2.03)
		0.3	7.96 ($\pm$ 0.26)	8.21 ($\pm$ 0.26)	2.64 ($\pm$ 0.83)	2.69 ($\pm$ 0.96)	32.40 ($\pm$ 2.47)	14.82 ($\pm$ 3.62)
		0.6	7.96 ($\pm$ 0.28)	8.00 ($\pm$ 0.27)	2.87 ($\pm$ 0.90)	2.92 ($\pm$ 1.03)	32.35 ($\pm$ 2.51)	30.51 ($\pm$ 3.66)
		0.9	7.97 ($\pm$ 0.30)	7.60 ($\pm$ 0.29)	3.09 ($\pm$ 0.96)	3.12 ($\pm$ 1.09)	32.35 ($\pm$ 2.59)	46.21 ($\pm$ 3.67)
	2000	0.0	6.52 ($\pm$ 0.13)	6.83 ($\pm$ 0.14)	1.58 ($\pm$ 0.56)	1.53 ($\pm$ 0.49)	40.76 ($\pm$ 2.52)	1.90 ($\pm$ 1.35)
		0.3	6.55 ($\pm$ 0.13)	6.80 ($\pm$ 0.13)	1.67 ($\pm$ 0.62)	1.61 ($\pm$ 0.58)	40.52 ($\pm$ 1.85)	13.24 ($\pm$ 2.74)
		0.6	6.55 ($\pm$ 0.13)	6.63 ($\pm$ 0.13)	1.80 ($\pm$ 0.65)	1.74 ($\pm$ 0.62)	40.56 ($\pm$ 1.81)	26.77 ($\pm$ 2.60)
		0.9	6.55 ($\pm$ 0.14)	6.29 ($\pm$ 0.13)	1.93 ($\pm$ 0.69)	1.85 ($\pm$ 0.64)	40.62 ($\pm$ 1.89)	40.36 ($\pm$ 2.34)
$\mathbf{A}_6(t)$	1000	0.0	6.26 ($\pm$ 0.22)	6.34 ($\pm$ 0.22)	2.08 ($\pm$ 0.86)	2.14 ($\pm$ 0.78)	22.13 ($\pm$ 2.88)	3.02 ($\pm$ 2.21)
		0.3	6.20 ($\pm$ 0.21)	6.19 ($\pm$ 0.20)	2.64 ($\pm$ 1.05)	2.55 ($\pm$ 1.01)	21.97 ($\pm$ 3.67)	9.59 ($\pm$ 3.29)
		0.6	6.19 ($\pm$ 0.23)	5.88 ($\pm$ 0.20)	2.86 ($\pm$ 1.15)	2.73 ($\pm$ 1.10)	21.94 ($\pm$ 3.87)	20.22 ($\pm$ 2.83)
		0.9	6.18 ($\pm$ 0.26)	5.26 ($\pm$ 0.21)	3.07 ($\pm$ 1.24)	2.88 ($\pm$ 1.14)	21.92 ($\pm$ 4.10)	30.87 ($\pm$ 2.19)
	2000	0.0	4.61 ($\pm$ 0.10)	4.68 ($\pm$ 0.10)	1.18 ($\pm$ 0.46)	1.15 ($\pm$ 0.44)	7.91 ($\pm$ 1.13)	1.58 ($\pm$ 1.01)
		0.3	4.64 ($\pm$ 0.14)	4.64 ($\pm$ 0.12)	1.54 ($\pm$ 0.64)	1.50 ($\pm$ 0.68)	8.20 ($\pm$ 1.19)	7.03 ($\pm$ 1.92)
		0.6	4.64 ($\pm$ 0.14)	4.42 ($\pm$ 0.11)	1.69 ($\pm$ 0.69)	1.62 ($\pm$ 0.74)	8.24 ($\pm$ 1.26)	14.72 ($\pm$ 1.77)
		0.9	4.63 ($\pm$ 0.14)	3.97 ($\pm$ 0.09)	1.84 ($\pm$ 0.73)	1.74 ($\pm$ 0.78)	8.31 ($\pm$ 1.40)	22.50 ($\pm$ 1.60)

Table 4: MAE errors ($\times 10^{-2}$) for Case 1(d) - $\sigma_3(t) = 0.5 + 0.5 \sin^2(\pi t)$.

function	n	ρ	MIAE		MAE		MIAE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
$\mathbf{A}_1(t)$	1000	0.0	5.17 ($\pm$ 0.18)	5.18 ($\pm$ 0.19)	2.08 ($\pm$ 0.79)	2.09 ($\pm$ 0.75)	31.65 ($\pm$ 2.68)	2.92 ($\pm$ 2.04)
		0.3	5.15 ($\pm$ 0.21)	5.03 ($\pm$ 0.18)	2.45 ($\pm$ 1.03)	2.44 ($\pm$ 1.05)	31.52 ($\pm$ 3.55)	7.70 ($\pm$ 3.10)
		0.6	5.16 ($\pm$ 0.24)	4.62 ($\pm$ 0.17)	2.68 ($\pm$ 1.10)	2.64 ($\pm$ 1.11)	31.47 ($\pm$ 3.79)	15.65 ($\pm$ 2.60)
		0.9	5.16 ($\pm$ 0.28)	3.67 ($\pm$ 0.19)	2.90 ($\pm$ 1.16)	2.79 ($\pm$ 1.12)	31.40 ($\pm$ 4.05)	23.60 ($\pm$ 2.02)
	2000	0.0	3.35 ($\pm$ 0.10)	3.35 ($\pm$ 0.10)	1.25 ($\pm$ 0.43)	1.23 ($\pm$ 0.43)	14.39 ($\pm$ 1.76)	1.31 ($\pm$ 0.81)
		0.3	3.38 ($\pm$ 0.11)	3.27 ($\pm$ 0.11)	1.38 ($\pm$ 0.59)	1.35 ($\pm$ 0.60)	14.46 ($\pm$ 1.71)	4.27 ($\pm$ 1.69)
		0.6	3.39 ($\pm$ 0.11)	2.94 ($\pm$ 0.10)	1.53 ($\pm$ 0.64)	1.46 ($\pm$ 0.66)	14.38 ($\pm$ 1.97)	8.81 ($\pm$ 1.48)
		0.9	3.39 ($\pm$ 0.11)	2.19 ($\pm$ 0.07)	1.67 ($\pm$ 0.69)	1.55 ($\pm$ 0.70)	14.32 ($\pm$ 2.29)	13.44 ($\pm$ 1.10)
$\mathbf{A}_2(t)$	1000	0.0	5.19 ($\pm$ 0.22)	5.20 ($\pm$ 0.22)	2.08 ($\pm$ 0.80)	2.11 ($\pm$ 0.76)	31.67 ($\pm$ 2.75)	3.07 ($\pm$ 2.20)
		0.3	5.13 ($\pm$ 0.18)	5.02 ($\pm$ 0.17)	2.46 ($\pm$ 1.00)	2.45 ($\pm$ 1.01)	31.56 ($\pm$ 3.61)	7.11 ($\pm$ 3.01)
		0.6	5.13 ($\pm$ 0.20)	4.60 ($\pm$ 0.16)	2.69 ($\pm$ 1.10)	2.65 ($\pm$ 1.10)	31.52 ($\pm$ 3.83)	15.04 ($\pm$ 2.38)
		0.9	5.13 ($\pm$ 0.23)	3.62 ($\pm$ 0.16)	2.92 ($\pm$ 1.16)	2.80 ($\pm$ 1.11)	31.47 ($\pm$ 4.09)	23.00 ($\pm$ 1.58)
	2000	0.0	3.34 ($\pm$ 0.08)	3.33 ($\pm$ 0.08)	1.20 ($\pm$ 0.46)	1.19 ($\pm$ 0.45)	14.82 ($\pm$ 1.71)	1.45 ($\pm$ 0.96)
		0.3	3.36 ($\pm$ 0.12)	3.25 ($\pm$ 0.12)	1.39 ($\pm$ 0.59)	1.35 ($\pm$ 0.63)	14.94 ($\pm$ 1.66)	4.05 ($\pm$ 1.61)
		0.6	3.36 ($\pm$ 0.12)	2.91 ($\pm$ 0.11)	1.54 ($\pm$ 0.64)	1.47 ($\pm$ 0.70)	14.87 ($\pm$ 1.93)	8.58 ($\pm$ 1.36)
		0.9	3.36 ($\pm$ 0.12)	2.12 ($\pm$ 0.08)	1.69 ($\pm$ 0.69)	1.58 ($\pm$ 0.74)	14.79 ($\pm$ 2.25)	13.19 ($\pm$ 1.04)
$\mathbf{A}_3(t)$	1000	0.0	6.60 ($\pm$ 0.23)	6.71 ($\pm$ 0.24)	2.21 ($\pm$ 0.77)	2.23 ($\pm$ 0.74)	19.12 ($\pm$ 2.45)	2.91 ($\pm$ 2.24)
		0.3	6.62 ($\pm$ 0.19)	6.66 ($\pm$ 0.17)	2.48 ($\pm$ 1.09)	2.49 ($\pm$ 1.09)	19.06 ($\pm$ 3.18)	11.24 ($\pm$ 3.14)
		0.6	6.63 ($\pm$ 0.20)	6.39 ($\pm$ 0.16)	2.72 ($\pm$ 1.16)	2.69 ($\pm$ 1.17)	19.13 ($\pm$ 3.23)	22.90 ($\pm$ 2.84)
		0.9	6.64 ($\pm$ 0.22)	5.86 ($\pm$ 0.16)	2.96 ($\pm$ 1.20)	2.85 ($\pm$ 1.23)	19.23 ($\pm$ 3.26)	34.59 ($\pm$ 2.55)
	2000	0.0	5.10 ($\pm$ 0.11)	5.19 ($\pm$ 0.12)	1.48 ($\pm$ 0.55)	1.47 ($\pm$ 0.57)	12.79 ($\pm$ 1.33)	1.30 ($\pm$ 0.97)
		0.3	5.11 ($\pm$ 0.12)	5.14 ($\pm$ 0.12)	1.40 ($\pm$ 0.60)	1.37 ($\pm$ 0.60)	13.23 ($\pm$ 1.41)	8.47 ($\pm$ 1.87)
		0.6	5.12 ($\pm$ 0.13)	4.94 ($\pm$ 0.11)	1.53 ($\pm$ 0.66)	1.48 ($\pm$ 0.64)	13.33 ($\pm$ 1.53)	17.14 ($\pm$ 1.83)
		0.9	5.12 ($\pm$ 0.13)	4.53 ($\pm$ 0.09)	1.65 ($\pm$ 0.72)	1.58 ($\pm$ 0.68)	13.45 ($\pm$ 1.70)	25.86 ($\pm$ 1.70)
$\mathbf{A}_4(t)$	1000	0.0	4.97 ($\pm$ 0.21)	4.98 ($\pm$ 0.21)	2.08 ($\pm$ 0.77)	2.10 ($\pm$ 0.72)	33.11 ($\pm$ 2.70)	2.93 ($\pm$ 2.14)
		0.3	4.94 ($\pm$ 0.19)	4.82 ($\pm$ 0.16)	2.43 ($\pm$ 1.03)	2.42 ($\pm$ 1.04)	32.96 ($\pm$ 3.54)	7.05 ($\pm$ 3.01)
		0.6	4.95 ($\pm$ 0.21)	4.37 ($\pm$ 0.15)	2.67 ($\pm$ 1.11)	2.61 ($\pm$ 1.13)	32.92 ($\pm$ 3.77)	14.53 ($\pm$ 2.42)
		0.9	4.95 ($\pm$ 0.25)	3.30 ($\pm$ 0.16)	2.90 ($\pm$ 1.16)	2.77 ($\pm$ 1.13)	32.86 ($\pm$ 4.03)	22.04 ($\pm$ 1.74)
	2000	0.0	3.06 ($\pm$ 0.09)	3.06 ($\pm$ 0.09)	1.26 ($\pm$ 0.44)	1.24 ($\pm$ 0.45)	16.05 ($\pm$ 1.84)	1.38 ($\pm$ 0.82)
		0.3	3.08 ($\pm$ 0.10)	2.96 ($\pm$ 0.10)	1.35 ($\pm$ 0.58)	1.32 ($\pm$ 0.59)	16.08 ($\pm$ 1.78)	3.71 ($\pm$ 1.48)
		0.6	3.08 ($\pm$ 0.09)	2.59 ($\pm$ 0.08)	1.50 ($\pm$ 0.64)	1.43 ($\pm$ 0.65)	15.99 ($\pm$ 2.05)	7.66 ($\pm$ 1.36)
		0.9	3.09 ($\pm$ 0.09)	1.66 ($\pm$ 0.06)	1.64 ($\pm$ 0.69)	1.53 ($\pm$ 0.69)	15.89 ($\pm$ 2.39)	11.73 ($\pm$ 0.95)
$\mathbf{A}_5(t)$	1000	0.0	7.26 ($\pm$ 0.19)	7.59 ($\pm$ 0.22)	2.34 ($\pm$ 0.93)	2.14 ($\pm$ 0.85)	25.26 ($\pm$ 2.29)	3.63 ($\pm$ 2.11)
		0.3	7.23 ($\pm$ 0.23)	7.52 ($\pm$ 0.24)	2.31 ($\pm$ 0.75)	2.33 ($\pm$ 0.83)	25.11 ($\pm$ 2.60)	15.73 ($\pm$ 3.80)
		0.6	7.23 ($\pm$ 0.25)	7.36 ($\pm$ 0.25)	2.51 ($\pm$ 0.78)	2.51 ($\pm$ 0.88)	25.06 ($\pm$ 2.68)	32.31 ($\pm$ 3.85)
		0.9	7.23 ($\pm$ 0.26)	7.08 ($\pm$ 0.26)	2.69 ($\pm$ 0.80)	2.65 ($\pm$ 0.96)	25.03 ($\pm$ 2.79)	48.89 ($\pm$ 3.85)
	2000	0.0	6.14 ($\pm$ 0.13)	6.49 ($\pm$ 0.14)	1.47 ($\pm$ 0.49)	1.33 ($\pm$ 0.44)	39.37 ($\pm$ 2.68)	2.28 ($\pm$ 1.74)
		0.3	6.16 ($\pm$ 0.12)	6.46 ($\pm$ 0.12)	1.52 ($\pm$ 0.58)	1.39 ($\pm$ 0.45)	38.85 ($\pm$ 2.12)	14.64 ($\pm$ 2.96)
		0.6	6.16 ($\pm$ 0.12)	6.34 ($\pm$ 0.12)	1.63 ($\pm$ 0.62)	1.49 ($\pm$ 0.48)	38.90 ($\pm$ 2.11)	29.59 ($\pm$ 2.81)
		0.9	6.15 ($\pm$ 0.13)	6.12 ($\pm$ 0.13)	1.74 ($\pm$ 0.66)	1.58 ($\pm$ 0.49)	39.00 ($\pm$ 2.17)	44.58 ($\pm$ 2.55)
$\mathbf{A}_6(t)$	1000	0.0	5.57 ($\pm$ 0.18)	5.63 ($\pm$ 0.18)	1.88 ($\pm$ 0.72)	1.85 ($\pm$ 0.69)	12.29 ($\pm$ 1.89)	2.83 ($\pm$ 2.15)
		0.3	5.52 ($\pm$ 0.18)	5.52 ($\pm$ 0.18)	2.27 ($\pm$ 0.86)	2.32 ($\pm$ 0.90)	12.39 ($\pm$ 2.44)	9.89 ($\pm$ 3.12)
		0.6	5.51 ($\pm$ 0.19)	5.28 ($\pm$ 0.18)	2.45 ($\pm$ 0.94)	2.47 ($\pm$ 0.98)	12.43 ($\pm$ 2.53)	20.64 ($\pm$ 2.74)
		0.9	5.49 ($\pm$ 0.21)	4.80 ($\pm$ 0.18)	2.63 ($\pm$ 0.98)	2.58 ($\pm$ 1.03)	12.48 ($\pm$ 2.66)	31.40 ($\pm$ 2.19)
	2000	0.0	4.29 ($\pm$ 0.09)	4.35 ($\pm$ 0.10)	1.04 ($\pm$ 0.40)	1.03 ($\pm$ 0.38)	9.57 ($\pm$ 1.10)	1.83 ($\pm$ 1.18)
		0.3	4.30 ($\pm$ 0.11)	4.32 ($\pm$ 0.10)	1.39 ($\pm$ 0.58)	1.36 ($\pm$ 0.56)	9.76 ($\pm$ 1.20)	7.66 ($\pm$ 1.92)
		0.6	4.30 ($\pm$ 0.11)	4.16 ($\pm$ 0.09)	1.52 ($\pm$ 0.63)	1.47 ($\pm$ 0.61)	9.77 ($\pm$ 1.36)	15.93 ($\pm$ 1.74)
		0.9	4.29 ($\pm$ 0.11)	3.83 ($\pm$ 0.07)	1.65 ($\pm$ 0.67)	1.58 ($\pm$ 0.65)	9.79 ($\pm$ 1.54)	24.27 ($\pm$ 1.57)

Below we present three examples, with fitted-versus-true curves for $\mathbf{A}_2$, $\mathbf{A}_4$, and $\mathbf{A}_6$, respectively. These plots show that our method is effective across a range of different $\mathbf{A}(t)$ configurations.

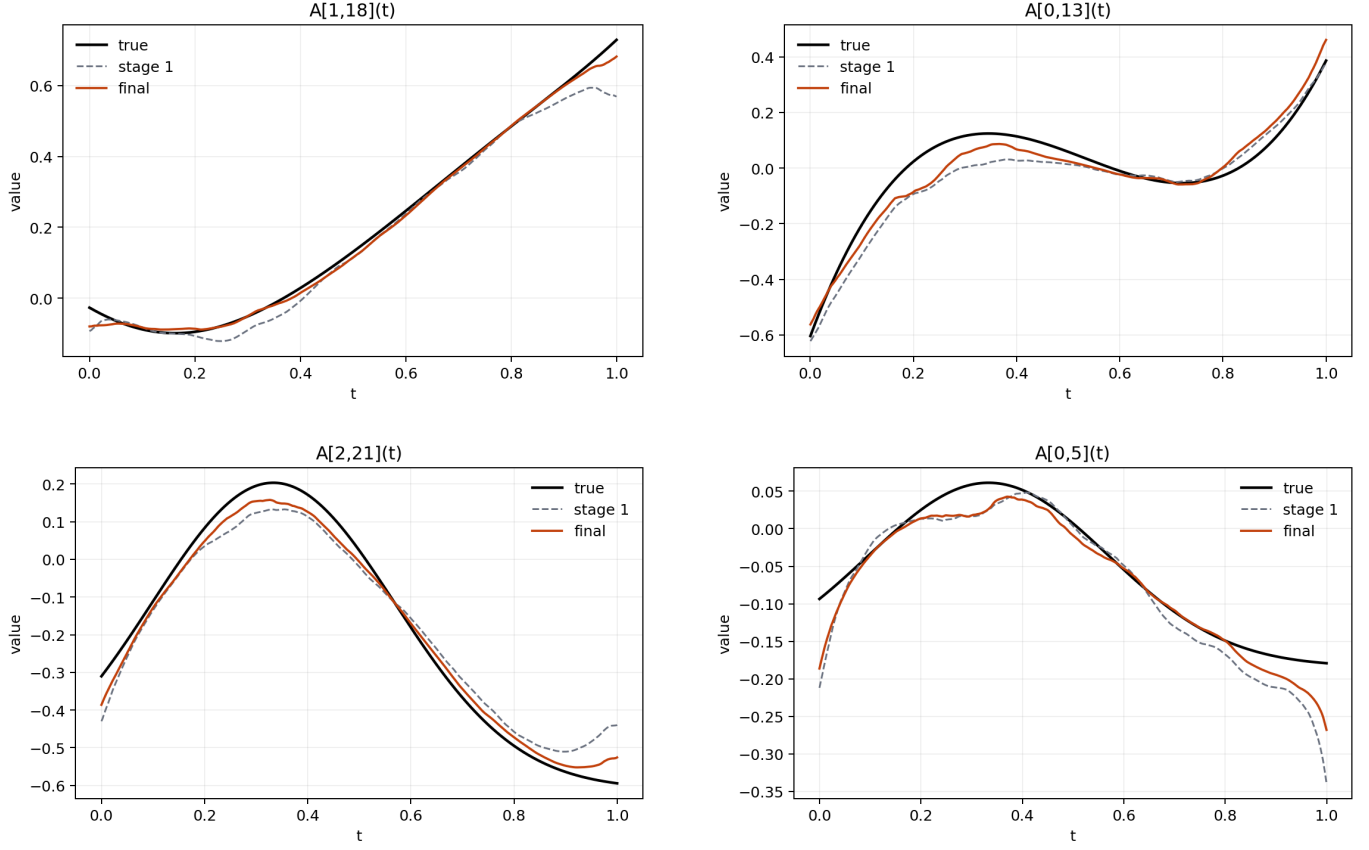


Figure 1: Selected estimated coefficient-function curves from Case 1.

We also present 3 examples of fitted-versus-true curves for $\sigma^2(t)$. They show that our method is effective for estimating the variance function across different settings.

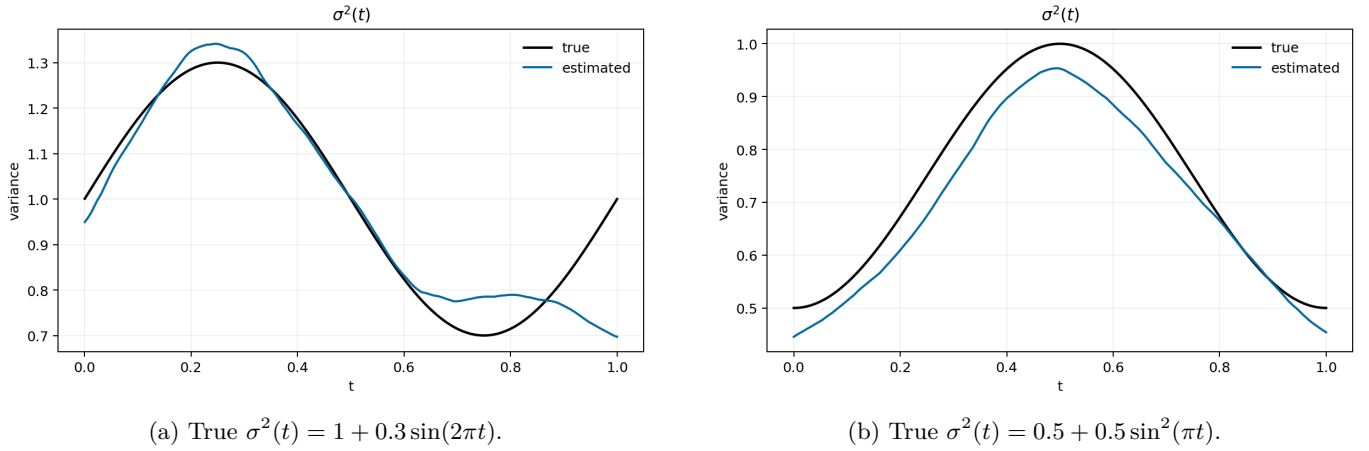


Figure 2: Selected estimated variance-function curves from Case 1.

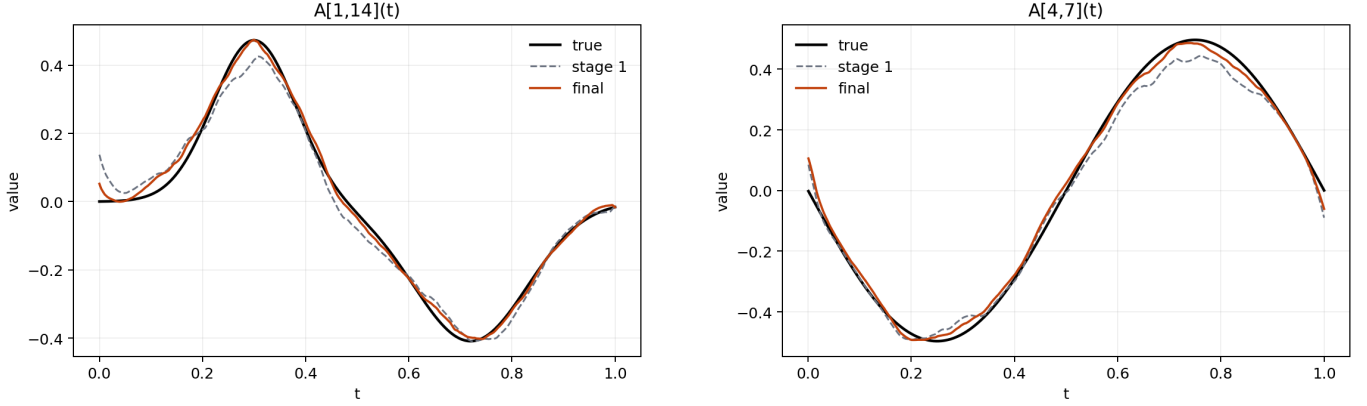
Case 2: 3D Tensor Covariate

DGP. The tensor size is set to be $D = 3$, $p_1 = p_2 = p_3 = 48$, and split by $p'_1 = p'_2 = p'_3 = 16$ such that by $S^{(1)} = S^{(2)} = S^{(3)} = 3$. Set the rank $R = 6$, and the s -th block tensor for j -th eye of i -th patient is generated by:

$$\mathcal{X}_{ij}^{(s)} = \sum_{r=1}^R x_{ijrs}^* \mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)} \circ \mathbf{x}_r^{(s3)}, \text{ where } x_{ijrs}^* \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1).$$

and any $\mathbf{x}_r^{(sd)}$ and $\mathbf{x}_{r'}^{(sd)}$ are orthonormal and obtained from the QR decomposition of a random matrix. The coefficient function is set to be $\mathcal{A}^{(s)}(t) = \sum_{r=1}^R \alpha_{rs}(t) \mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)} \circ \mathbf{x}_r^{(s3)}$ where $\alpha_{rs}(t)$ has the same 4 choices as in Case 1. And everything else is the same.

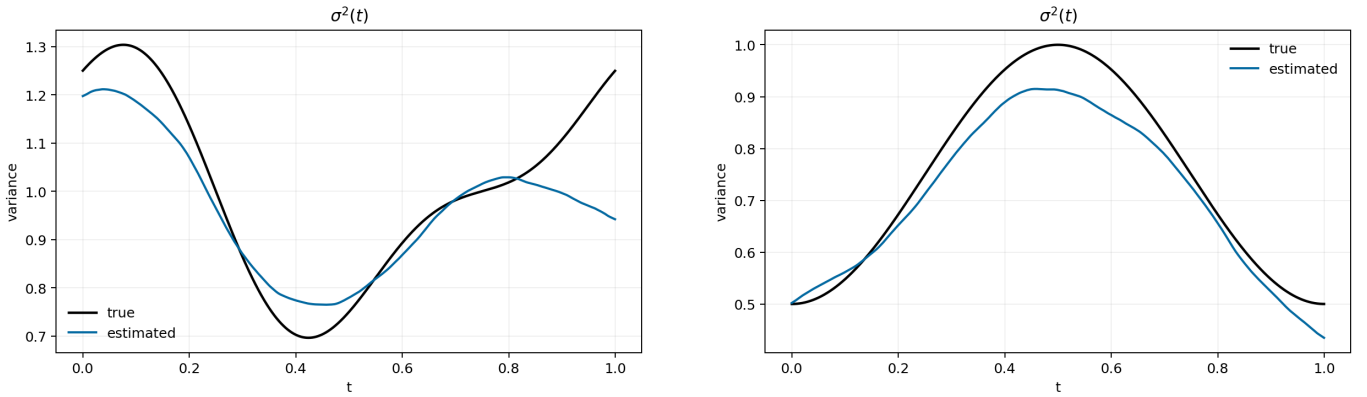
Simulation Results. We vary $n = 2000/5000$, coefficients $\alpha_{1rs}(t), \dots, \alpha_{6rs}(t)$, and correlation $\rho = 0/0.3/0.6/0.9$. Tables 5–8 summarize the Case 2 results under the four variance settings. Figure 3 presents representative fitted-versus-true coefficient-function curves under $\mathbf{A}_5(t)$ and $\mathbf{A}_3(t)$, and Figure 4 presents two representative fitted-versus-true variance-function curves.



(a) Coefficient function under $\mathbf{A}_5(t)$.

(b) Coefficient function under $\mathbf{A}_3(t)$.

Figure 3: Representative fitted-versus-true coefficient-function curves from Case 2.



(a) $\sigma^2(t) = 1 + 0.25 \cos(2\pi t) + 0.1 \sin(4\pi t)$.

(b) $\sigma^2(t) = 0.5 + 0.5 \sin^2(\pi t)$.

Figure 4: Representative fitted-versus-true variance-function curves from Case 2.

Table 5: MAE errors ($\times 10^{-2}$) for Case 2(a) - $\sigma_0^2(t) \equiv 1$.

function	n	ρ	MIAE		MAE		MIAE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
$\mathbf{A}_1(t)$	2000	0.0	3.51 ($\pm$ 0.10)	3.53 ($\pm$ 0.10)	1.55 ($\pm$ 0.65)	1.53 ($\pm$ 0.66)	18.86 ($\pm$ 2.34)	1.96 ($\pm$ 1.45)
		0.3	3.51 ($\pm$ 0.10)	3.44 ($\pm$ 0.10)	1.70 ($\pm$ 0.71)	1.69 ($\pm$ 0.71)	18.64 ($\pm$ 2.18)	6.54 ($\pm$ 2.61)
		0.6	3.51 ($\pm$ 0.10)	3.17 ($\pm$ 0.09)	1.86 ($\pm$ 0.77)	1.83 ($\pm$ 0.75)	18.61 ($\pm$ 2.62)	13.45 ($\pm$ 2.06)
		0.9	3.52 ($\pm$ 0.11)	2.58 ($\pm$ 0.08)	2.02 ($\pm$ 0.83)	1.96 ($\pm$ 0.77)	18.66 ($\pm$ 2.97)	20.39 ($\pm$ 1.34)
	5000	0.0	2.42 ($\pm$ 0.06)	2.43 ($\pm$ 0.06)	0.90 ($\pm$ 0.36)	0.91 ($\pm$ 0.37)	4.58 ($\pm$ 0.92)	1.15 ($\pm$ 0.78)
		0.3	2.39 ($\pm$ 0.06)	2.35 ($\pm$ 0.06)	0.95 ($\pm$ 0.30)	0.94 ($\pm$ 0.30)	4.61 ($\pm$ 0.98)	3.99 ($\pm$ 1.29)
		0.6	2.39 ($\pm$ 0.06)	2.20 ($\pm$ 0.06)	1.04 ($\pm$ 0.34)	1.03 ($\pm$ 0.32)	4.78 ($\pm$ 1.16)	7.73 ($\pm$ 1.04)
		0.9	2.39 ($\pm$ 0.06)	1.89 ($\pm$ 0.04)	1.14 ($\pm$ 0.37)	1.11 ($\pm$ 0.34)	5.03 ($\pm$ 1.35)	11.42 ($\pm$ 0.64)
$\mathbf{A}_2(t)$	2000	0.0	3.46 ($\pm$ 0.10)	3.48 ($\pm$ 0.10)	1.52 ($\pm$ 0.72)	1.50 ($\pm$ 0.73)	19.09 ($\pm$ 2.34)	1.68 ($\pm$ 1.37)
		0.3	3.45 ($\pm$ 0.09)	3.39 ($\pm$ 0.10)	1.75 ($\pm$ 0.68)	1.73 ($\pm$ 0.67)	18.90 ($\pm$ 2.18)	6.28 ($\pm$ 2.39)
		0.6	3.45 ($\pm$ 0.09)	3.11 ($\pm$ 0.09)	1.91 ($\pm$ 0.74)	1.87 ($\pm$ 0.74)	18.89 ($\pm$ 2.55)	13.04 ($\pm$ 2.10)
		0.9	3.45 ($\pm$ 0.10)	2.47 ($\pm$ 0.08)	2.06 ($\pm$ 0.80)	1.99 ($\pm$ 0.78)	18.92 ($\pm$ 2.91)	19.93 ($\pm$ 1.51)
	5000	0.0	2.35 ($\pm$ 0.05)	2.36 ($\pm$ 0.05)	0.91 ($\pm$ 0.38)	0.91 ($\pm$ 0.39)	6.60 ($\pm$ 0.93)	1.21 ($\pm$ 0.73)
		0.3	2.33 ($\pm$ 0.05)	2.29 ($\pm$ 0.05)	0.92 ($\pm$ 0.30)	0.91 ($\pm$ 0.29)	6.61 ($\pm$ 0.96)	3.91 ($\pm$ 1.23)
		0.6	2.32 ($\pm$ 0.05)	2.12 ($\pm$ 0.04)	1.02 ($\pm$ 0.32)	1.00 ($\pm$ 0.31)	6.73 ($\pm$ 1.09)	7.63 ($\pm$ 0.99)
		0.9	2.32 ($\pm$ 0.05)	1.74 ($\pm$ 0.04)	1.12 ($\pm$ 0.36)	1.08 ($\pm$ 0.33)	6.88 ($\pm$ 1.27)	11.32 ($\pm$ 0.66)
$\mathbf{A}_3(t)$	2000	0.0	4.56 ($\pm$ 0.11)	4.60 ($\pm$ 0.11)	1.55 ($\pm$ 0.72)	1.55 ($\pm$ 0.71)	22.37 ($\pm$ 2.16)	1.89 ($\pm$ 1.60)
		0.3	4.56 ($\pm$ 0.11)	4.52 ($\pm$ 0.10)	1.76 ($\pm$ 0.79)	1.75 ($\pm$ 0.82)	22.36 ($\pm$ 2.18)	9.36 ($\pm$ 2.40)
		0.6	4.62 ($\pm$ 0.12)	4.25 ($\pm$ 0.10)	1.93 ($\pm$ 0.85)	1.89 ($\pm$ 0.91)	31.24 ($\pm$ 2.40)	18.48 ($\pm$ 1.71)
		0.9	4.62 ($\pm$ 0.13)	3.63 ($\pm$ 0.10)	2.06 ($\pm$ 0.90)	1.99 ($\pm$ 0.94)	31.17 ($\pm$ 2.80)	28.21 ($\pm$ 1.02)
	5000	0.0	2.70 ($\pm$ 0.05)	2.70 ($\pm$ 0.05)	0.93 ($\pm$ 0.31)	0.93 ($\pm$ 0.33)	13.81 ($\pm$ 1.23)	1.10 ($\pm$ 0.68)
		0.3	2.68 ($\pm$ 0.06)	2.61 ($\pm$ 0.05)	0.95 ($\pm$ 0.35)	0.94 ($\pm$ 0.35)	13.72 ($\pm$ 1.23)	4.90 ($\pm$ 1.17)
		0.6	2.68 ($\pm$ 0.06)	2.36 ($\pm$ 0.05)	1.06 ($\pm$ 0.37)	1.02 ($\pm$ 0.37)	13.78 ($\pm$ 1.33)	9.50 ($\pm$ 0.95)
		0.9	2.78 ($\pm$ 0.06)	1.68 ($\pm$ 0.04)	1.16 ($\pm$ 0.39)	1.10 ($\pm$ 0.36)	19.45 ($\pm$ 1.49)	14.72 ($\pm$ 0.59)
$\mathbf{A}_4(t)$	2000	0.0	3.13 ($\pm$ 0.09)	3.14 ($\pm$ 0.09)	1.51 ($\pm$ 0.67)	1.49 ($\pm$ 0.67)	22.56 ($\pm$ 2.39)	1.83 ($\pm$ 1.46)
		0.3	3.12 ($\pm$ 0.08)	3.03 ($\pm$ 0.08)	1.68 ($\pm$ 0.70)	1.67 ($\pm$ 0.71)	22.38 ($\pm$ 2.22)	5.44 ($\pm$ 2.37)
		0.6	3.13 ($\pm$ 0.09)	2.71 ($\pm$ 0.07)	1.85 ($\pm$ 0.77)	1.81 ($\pm$ 0.77)	22.33 ($\pm$ 2.70)	11.14 ($\pm$ 1.88)
		0.9	3.13 ($\pm$ 0.10)	1.91 ($\pm$ 0.06)	2.00 ($\pm$ 0.82)	1.94 ($\pm$ 0.80)	22.29 ($\pm$ 3.24)	16.95 ($\pm$ 1.02)
	5000	0.0	1.87 ($\pm$ 0.04)	1.87 ($\pm$ 0.04)	0.91 ($\pm$ 0.33)	0.91 ($\pm$ 0.34)	8.40 ($\pm$ 1.33)	1.13 ($\pm$ 0.74)
		0.3	1.84 ($\pm$ 0.05)	1.78 ($\pm$ 0.05)	0.93 ($\pm$ 0.30)	0.92 ($\pm$ 0.30)	8.34 ($\pm$ 1.30)	2.81 ($\pm$ 1.20)
		0.6	1.84 ($\pm$ 0.06)	1.57 ($\pm$ 0.05)	1.03 ($\pm$ 0.33)	1.01 ($\pm$ 0.32)	8.43 ($\pm$ 1.43)	5.30 ($\pm$ 0.92)
		0.9	1.84 ($\pm$ 0.06)	1.06 ($\pm$ 0.03)	1.13 ($\pm$ 0.37)	1.10 ($\pm$ 0.34)	8.52 ($\pm$ 1.65)	7.72 ($\pm$ 0.48)
$\mathbf{A}_5(t)$	2000	0.0	5.38 ($\pm$ 0.13)	5.41 ($\pm$ 0.13)	1.69 ($\pm$ 0.65)	1.69 ($\pm$ 0.70)	22.85 ($\pm$ 1.37)	2.04 ($\pm$ 1.43)
		0.3	5.39 ($\pm$ 0.15)	5.36 ($\pm$ 0.15)	1.95 ($\pm$ 0.86)	1.88 ($\pm$ 0.93)	22.74 ($\pm$ 1.56)	11.82 ($\pm$ 2.49)
		0.6	5.39 ($\pm$ 0.16)	5.12 ($\pm$ 0.16)	2.09 ($\pm$ 0.91)	2.01 ($\pm$ 1.01)	22.77 ($\pm$ 1.78)	24.36 ($\pm$ 2.28)
		0.9	5.39 ($\pm$ 0.18)	4.66 ($\pm$ 0.17)	2.21 ($\pm$ 0.95)	2.11 ($\pm$ 1.05)	22.82 ($\pm$ 2.03)	36.95 ($\pm$ 2.04)
	5000	0.0	3.13 ($\pm$ 0.05)	3.14 ($\pm$ 0.05)	0.94 ($\pm$ 0.42)	0.95 ($\pm$ 0.43)	15.82 ($\pm$ 1.17)	1.26 ($\pm$ 0.66)
		0.3	3.11 ($\pm$ 0.05)	3.04 ($\pm$ 0.05)	1.00 ($\pm$ 0.29)	0.99 ($\pm$ 0.30)	15.76 ($\pm$ 1.25)	6.00 ($\pm$ 1.24)
		0.6	3.11 ($\pm$ 0.06)	2.77 ($\pm$ 0.04)	1.09 ($\pm$ 0.32)	1.07 ($\pm$ 0.31)	15.80 ($\pm$ 1.34)	11.97 ($\pm$ 1.03)
		0.9	3.10 ($\pm$ 0.06)	2.16 ($\pm$ 0.03)	1.19 ($\pm$ 0.35)	1.14 ($\pm$ 0.33)	15.84 ($\pm$ 1.49)	17.89 ($\pm$ 0.72)
$\mathbf{A}_6(t)$	2000	0.0	4.42 ($\pm$ 0.10)	4.44 ($\pm$ 0.11)	1.60 ($\pm$ 0.77)	1.62 ($\pm$ 0.78)	24.10 ($\pm$ 2.05)	1.86 ($\pm$ 1.08)
		0.3	4.42 ($\pm$ 0.11)	4.36 ($\pm$ 0.12)	1.76 ($\pm$ 0.73)	1.74 ($\pm$ 0.74)	24.05 ($\pm$ 2.19)	8.50 ($\pm$ 2.63)
		0.6	4.42 ($\pm$ 0.11)	4.10 ($\pm$ 0.11)	1.92 ($\pm$ 0.79)	1.88 ($\pm$ 0.80)	23.99 ($\pm$ 2.60)	17.91 ($\pm$ 2.19)
		0.9	4.55 ($\pm$ 0.13)	3.51 ($\pm$ 0.11)	2.05 ($\pm$ 0.87)	1.97 ($\pm$ 0.88)	31.82 ($\pm$ 2.90)	27.33 ($\pm$ 1.36)
	5000	0.0	2.67 ($\pm$ 0.05)	2.67 ($\pm$ 0.05)	0.93 ($\pm$ 0.35)	0.92 ($\pm$ 0.35)	14.08 ($\pm$ 1.23)	1.17 ($\pm$ 0.77)
		0.3	2.65 ($\pm$ 0.05)	2.58 ($\pm$ 0.05)	0.97 ($\pm$ 0.31)	0.95 ($\pm$ 0.32)	14.02 ($\pm$ 1.25)	4.64 ($\pm$ 1.18)
		0.6	2.65 ($\pm$ 0.05)	2.33 ($\pm$ 0.05)	1.07 ($\pm$ 0.34)	1.03 ($\pm$ 0.33)	14.07 ($\pm$ 1.36)	9.18 ($\pm$ 0.97)
		0.9	2.78 ($\pm$ 0.06)	1.67 ($\pm$ 0.04)	1.17 ($\pm$ 0.36)	1.10 ($\pm$ 0.35)	19.51 ($\pm$ 1.51)	14.59 ($\pm$ 0.55)

Table 6: MAE errors ($\times 10^{-2}$) for Case 2(b) - $\sigma_1^2(t) = 1 + 0.3 \sin(2\pi t)$.

function	n	ρ	MIAE		MAE		MIAE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
$\mathbf{A}_1(t)$	2000	0.0	3.52 ($\pm$ 0.10)	3.52 ($\pm$ 0.10)	1.54 ($\pm$ 0.66)	1.52 ($\pm$ 0.66)	22.39 ($\pm$ 2.22)	1.96 ($\pm$ 1.43)
		0.3	3.51 ($\pm$ 0.09)	3.43 ($\pm$ 0.09)	1.72 ($\pm$ 0.69)	1.63 ($\pm$ 0.70)	22.33 ($\pm$ 2.07)	6.34 ($\pm$ 2.58)
		0.6	3.51 ($\pm$ 0.10)	3.13 ($\pm$ 0.09)	1.88 ($\pm$ 0.74)	1.76 ($\pm$ 0.77)	22.27 ($\pm$ 2.51)	13.21 ($\pm$ 1.97)
		0.9	3.52 ($\pm$ 0.11)	2.46 ($\pm$ 0.08)	2.02 ($\pm$ 0.80)	1.89 ($\pm$ 0.78)	22.23 ($\pm$ 2.92)	20.14 ($\pm$ 1.26)
	5000	0.0	2.28 ($\pm$ 0.06)	2.28 ($\pm$ 0.06)	0.91 ($\pm$ 0.36)	0.90 ($\pm$ 0.37)	7.11 ($\pm$ 1.15)	1.16 ($\pm$ 0.78)
		0.3	2.24 ($\pm$ 0.06)	2.19 ($\pm$ 0.06)	0.93 ($\pm$ 0.30)	0.92 ($\pm$ 0.29)	7.10 ($\pm$ 1.21)	3.62 ($\pm$ 1.23)
		0.6	2.24 ($\pm$ 0.06)	2.02 ($\pm$ 0.05)	1.03 ($\pm$ 0.33)	1.01 ($\pm$ 0.31)	7.20 ($\pm$ 1.32)	7.03 ($\pm$ 0.99)
		0.9	2.24 ($\pm$ 0.06)	1.64 ($\pm$ 0.04)	1.12 ($\pm$ 0.36)	1.10 ($\pm$ 0.33)	7.36 ($\pm$ 1.50)	10.39 ($\pm$ 0.61)
$\mathbf{A}_2(t)$	2000	0.0	3.46 ($\pm$ 0.10)	3.48 ($\pm$ 0.10)	1.50 ($\pm$ 0.70)	1.47 ($\pm$ 0.70)	22.57 ($\pm$ 2.21)	1.72 ($\pm$ 1.38)
		0.3	3.46 ($\pm$ 0.10)	3.39 ($\pm$ 0.10)	1.74 ($\pm$ 0.66)	1.64 ($\pm$ 0.69)	22.53 ($\pm$ 2.07)	6.13 ($\pm$ 2.33)
		0.6	3.46 ($\pm$ 0.10)	3.09 ($\pm$ 0.09)	1.91 ($\pm$ 0.71)	1.78 ($\pm$ 0.76)	22.45 ($\pm$ 2.52)	12.87 ($\pm$ 2.01)
		0.9	3.46 ($\pm$ 0.11)	2.39 ($\pm$ 0.08)	2.06 ($\pm$ 0.77)	1.90 ($\pm$ 0.80)	22.39 ($\pm$ 2.98)	19.77 ($\pm$ 1.44)
	5000	0.0	2.22 ($\pm$ 0.05)	2.23 ($\pm$ 0.05)	0.91 ($\pm$ 0.36)	0.90 ($\pm$ 0.38)	6.84 ($\pm$ 1.25)	1.25 ($\pm$ 0.73)
		0.3	2.20 ($\pm$ 0.05)	2.16 ($\pm$ 0.04)	0.91 ($\pm$ 0.31)	0.90 ($\pm$ 0.29)	6.79 ($\pm$ 1.18)	3.56 ($\pm$ 1.18)
		0.6	2.19 ($\pm$ 0.05)	1.97 ($\pm$ 0.04)	1.01 ($\pm$ 0.34)	0.99 ($\pm$ 0.31)	6.90 ($\pm$ 1.29)	6.97 ($\pm$ 0.94)
		0.9	2.19 ($\pm$ 0.05)	1.54 ($\pm$ 0.03)	1.11 ($\pm$ 0.36)	1.07 ($\pm$ 0.32)	7.05 ($\pm$ 1.46)	10.34 ($\pm$ 0.62)
$\mathbf{A}_3(t)$	2000	0.0	4.56 ($\pm$ 0.11)	4.58 ($\pm$ 0.11)	1.55 ($\pm$ 0.70)	1.50 ($\pm$ 0.71)	22.24 ($\pm$ 2.16)	1.97 ($\pm$ 1.52)
		0.3	4.56 ($\pm$ 0.10)	4.51 ($\pm$ 0.10)	1.77 ($\pm$ 0.77)	1.72 ($\pm$ 0.82)	22.39 ($\pm$ 2.12)	9.33 ($\pm$ 2.34)
		0.6	4.61 ($\pm$ 0.12)	4.24 ($\pm$ 0.10)	1.94 ($\pm$ 0.83)	1.85 ($\pm$ 0.89)	31.19 ($\pm$ 2.41)	18.46 ($\pm$ 1.65)
		0.9	4.62 ($\pm$ 0.13)	3.62 ($\pm$ 0.09)	2.07 ($\pm$ 0.87)	1.94 ($\pm$ 0.92)	31.08 ($\pm$ 2.79)	28.22 ($\pm$ 1.05)
	5000	0.0	2.69 ($\pm$ 0.05)	2.69 ($\pm$ 0.05)	0.93 ($\pm$ 0.31)	0.92 ($\pm$ 0.34)	13.72 ($\pm$ 1.22)	1.15 ($\pm$ 0.76)
		0.3	2.68 ($\pm$ 0.06)	2.61 ($\pm$ 0.05)	0.95 ($\pm$ 0.35)	0.93 ($\pm$ 0.34)	13.66 ($\pm$ 1.25)	4.84 ($\pm$ 1.14)
		0.6	2.68 ($\pm$ 0.06)	2.36 ($\pm$ 0.05)	1.05 ($\pm$ 0.38)	1.01 ($\pm$ 0.35)	13.71 ($\pm$ 1.34)	9.45 ($\pm$ 0.94)
		0.9	2.77 ($\pm$ 0.06)	1.67 ($\pm$ 0.04)	1.16 ($\pm$ 0.39)	1.08 ($\pm$ 0.35)	19.35 ($\pm$ 1.49)	14.69 ($\pm$ 0.59)
$\mathbf{A}_4(t)$	2000	0.0	3.25 ($\pm$ 0.09)	3.26 ($\pm$ 0.09)	1.49 ($\pm$ 0.68)	1.48 ($\pm$ 0.67)	24.79 ($\pm$ 2.24)	1.84 ($\pm$ 1.49)
		0.3	3.25 ($\pm$ 0.08)	3.15 ($\pm$ 0.08)	1.70 ($\pm$ 0.69)	1.61 ($\pm$ 0.72)	24.76 ($\pm$ 2.11)	5.60 ($\pm$ 2.34)
		0.6	3.25 ($\pm$ 0.09)	2.82 ($\pm$ 0.07)	1.86 ($\pm$ 0.75)	1.75 ($\pm$ 0.78)	24.68 ($\pm$ 2.57)	11.64 ($\pm$ 1.82)
		0.9	3.26 ($\pm$ 0.10)	2.00 ($\pm$ 0.06)	2.01 ($\pm$ 0.80)	1.88 ($\pm$ 0.81)	24.61 ($\pm$ 3.06)	17.83 ($\pm$ 1.03)
	5000	0.0	1.89 ($\pm$ 0.05)	1.89 ($\pm$ 0.04)	0.91 ($\pm$ 0.33)	0.90 ($\pm$ 0.34)	9.48 ($\pm$ 1.31)	1.20 ($\pm$ 0.72)
		0.3	1.86 ($\pm$ 0.05)	1.80 ($\pm$ 0.05)	0.92 ($\pm$ 0.31)	0.91 ($\pm$ 0.29)	9.44 ($\pm$ 1.26)	2.82 ($\pm$ 1.16)
		0.6	1.86 ($\pm$ 0.06)	1.57 ($\pm$ 0.05)	1.02 ($\pm$ 0.34)	1.00 ($\pm$ 0.31)	9.53 ($\pm$ 1.37)	5.38 ($\pm$ 0.90)
		0.9	1.86 ($\pm$ 0.06)	1.02 ($\pm$ 0.03)	1.11 ($\pm$ 0.37)	1.08 ($\pm$ 0.32)	9.64 ($\pm$ 1.55)	7.89 ($\pm$ 0.50)
$\mathbf{A}_5(t)$	2000	0.0	5.34 ($\pm$ 0.12)	5.38 ($\pm$ 0.13)	1.69 ($\pm$ 0.63)	1.64 ($\pm$ 0.68)	20.41 ($\pm$ 1.81)	1.99 ($\pm$ 1.48)
		0.3	5.36 ($\pm$ 0.15)	5.32 ($\pm$ 0.15)	1.96 ($\pm$ 0.82)	1.81 ($\pm$ 0.90)	20.48 ($\pm$ 2.03)	11.74 ($\pm$ 2.52)
		0.6	5.36 ($\pm$ 0.16)	5.09 ($\pm$ 0.16)	2.10 ($\pm$ 0.87)	1.94 ($\pm$ 0.98)	20.44 ($\pm$ 2.30)	24.30 ($\pm$ 2.31)
		0.9	5.36 ($\pm$ 0.18)	4.63 ($\pm$ 0.16)	2.22 ($\pm$ 0.91)	2.03 ($\pm$ 1.03)	20.40 ($\pm$ 2.57)	36.92 ($\pm$ 2.09)
	5000	0.0	3.11 ($\pm$ 0.05)	3.11 ($\pm$ 0.05)	0.94 ($\pm$ 0.40)	0.93 ($\pm$ 0.43)	15.72 ($\pm$ 1.16)	1.31 ($\pm$ 0.71)
		0.3	3.09 ($\pm$ 0.05)	3.02 ($\pm$ 0.05)	0.98 ($\pm$ 0.30)	0.97 ($\pm$ 0.31)	15.70 ($\pm$ 1.29)	5.98 ($\pm$ 1.21)
		0.6	3.08 ($\pm$ 0.05)	2.75 ($\pm$ 0.04)	1.08 ($\pm$ 0.33)	1.05 ($\pm$ 0.32)	15.72 ($\pm$ 1.37)	11.95 ($\pm$ 1.01)
		0.9	3.08 ($\pm$ 0.06)	2.15 ($\pm$ 0.03)	1.17 ($\pm$ 0.36)	1.12 ($\pm$ 0.33)	15.75 ($\pm$ 1.50)	17.89 ($\pm$ 0.70)
$\mathbf{A}_6(t)$	2000	0.0	4.40 ($\pm$ 0.10)	4.42 ($\pm$ 0.10)	1.61 ($\pm$ 0.74)	1.61 ($\pm$ 0.76)	23.98 ($\pm$ 2.01)	1.83 ($\pm$ 1.08)
		0.3	4.40 ($\pm$ 0.11)	4.35 ($\pm$ 0.12)	1.79 ($\pm$ 0.70)	1.68 ($\pm$ 0.72)	24.06 ($\pm$ 2.15)	8.40 ($\pm$ 2.57)
		0.6	4.40 ($\pm$ 0.11)	4.08 ($\pm$ 0.11)	1.94 ($\pm$ 0.76)	1.81 ($\pm$ 0.79)	23.96 ($\pm$ 2.56)	17.82 ($\pm$ 2.12)
		0.9	4.53 ($\pm$ 0.13)	3.50 ($\pm$ 0.11)	2.08 ($\pm$ 0.84)	1.90 ($\pm$ 0.87)	31.74 ($\pm$ 2.88)	27.29 ($\pm$ 1.34)
	5000	0.0	2.66 ($\pm$ 0.05)	2.66 ($\pm$ 0.05)	0.93 ($\pm$ 0.34)	0.91 ($\pm$ 0.36)	13.99 ($\pm$ 1.23)	1.23 ($\pm$ 0.78)
		0.3	2.64 ($\pm$ 0.05)	2.57 ($\pm$ 0.05)	0.96 ($\pm$ 0.31)	0.94 ($\pm$ 0.31)	13.96 ($\pm$ 1.28)	4.60 ($\pm$ 1.14)
		0.6	2.64 ($\pm$ 0.05)	2.32 ($\pm$ 0.05)	1.06 ($\pm$ 0.34)	1.02 ($\pm$ 0.32)	14.00 ($\pm$ 1.38)	9.15 ($\pm$ 0.92)
		0.9	2.76 ($\pm$ 0.06)	1.66 ($\pm$ 0.04)	1.16 ($\pm$ 0.37)	1.09 ($\pm$ 0.34)	19.41 ($\pm$ 1.51)	14.58 ($\pm$ 0.51)

Table 7: MAE errors ($\times 10^{-2}$) for Case 2(c) - $\sigma_2^2(t) = 1 + 0.25 \cos(2\pi t) + 0.1 \sin(4\pi t)$.

function	n	ρ	MIAE		MAE		MIAE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
$\mathbf{A}_1(t)$	2000	0.0	3.60 ($\pm$ 0.09)	3.60 ($\pm$ 0.09)	1.51 ($\pm$ 0.68)	1.52 ($\pm$ 0.68)	23.71 ($\pm$ 2.11)	1.91 ($\pm$ 1.38)
		0.3	3.59 ($\pm$ 0.10)	3.50 ($\pm$ 0.10)	1.70 ($\pm$ 0.71)	1.67 ($\pm$ 0.70)	23.57 ($\pm$ 1.96)	6.71 ($\pm$ 2.51)
		0.6	3.59 ($\pm$ 0.10)	3.20 ($\pm$ 0.09)	1.87 ($\pm$ 0.77)	1.81 ($\pm$ 0.76)	23.57 ($\pm$ 2.36)	13.85 ($\pm$ 1.93)
		0.9	3.59 ($\pm$ 0.11)	2.53 ($\pm$ 0.08)	2.02 ($\pm$ 0.82)	1.93 ($\pm$ 0.78)	23.61 ($\pm$ 2.78)	21.00 ($\pm$ 1.23)
	5000	0.0	2.32 ($\pm$ 0.06)	2.31 ($\pm$ 0.06)	0.92 ($\pm$ 0.37)	0.89 ($\pm$ 0.36)	8.74 ($\pm$ 1.05)	1.06 ($\pm$ 0.74)
		0.3	2.29 ($\pm$ 0.06)	2.23 ($\pm$ 0.06)	0.94 ($\pm$ 0.30)	0.93 ($\pm$ 0.29)	8.81 ($\pm$ 1.05)	3.82 ($\pm$ 1.32)
		0.6	2.28 ($\pm$ 0.06)	2.05 ($\pm$ 0.05)	1.04 ($\pm$ 0.33)	1.02 ($\pm$ 0.31)	8.95 ($\pm$ 1.24)	7.32 ($\pm$ 1.06)
		0.9	2.28 ($\pm$ 0.06)	1.66 ($\pm$ 0.04)	1.14 ($\pm$ 0.37)	1.10 ($\pm$ 0.33)	9.11 ($\pm$ 1.49)	10.76 ($\pm$ 0.63)
$\mathbf{A}_2(t)$	2000	0.0	3.56 ($\pm$ 0.10)	3.57 ($\pm$ 0.10)	1.48 ($\pm$ 0.73)	1.49 ($\pm$ 0.72)	23.95 ($\pm$ 2.07)	1.66 ($\pm$ 1.32)
		0.3	3.56 ($\pm$ 0.10)	3.47 ($\pm$ 0.10)	1.73 ($\pm$ 0.69)	1.70 ($\pm$ 0.69)	23.82 ($\pm$ 1.90)	6.48 ($\pm$ 2.41)
		0.6	3.56 ($\pm$ 0.10)	3.17 ($\pm$ 0.09)	1.89 ($\pm$ 0.75)	1.83 ($\pm$ 0.75)	23.85 ($\pm$ 2.25)	13.54 ($\pm$ 1.96)
		0.9	3.56 ($\pm$ 0.11)	2.46 ($\pm$ 0.08)	2.05 ($\pm$ 0.80)	1.95 ($\pm$ 0.80)	23.91 ($\pm$ 2.65)	20.66 ($\pm$ 1.36)
	5000	0.0	2.28 ($\pm$ 0.06)	2.27 ($\pm$ 0.05)	0.92 ($\pm$ 0.38)	0.90 ($\pm$ 0.37)	10.41 ($\pm$ 1.04)	1.15 ($\pm$ 0.71)
		0.3	2.26 ($\pm$ 0.05)	2.20 ($\pm$ 0.04)	0.92 ($\pm$ 0.31)	0.92 ($\pm$ 0.28)	10.45 ($\pm$ 1.07)	3.76 ($\pm$ 1.24)
		0.6	2.26 ($\pm$ 0.05)	2.00 ($\pm$ 0.04)	1.02 ($\pm$ 0.34)	1.00 ($\pm$ 0.31)	10.56 ($\pm$ 1.24)	7.25 ($\pm$ 0.99)
		0.9	2.25 ($\pm$ 0.05)	1.55 ($\pm$ 0.03)	1.12 ($\pm$ 0.36)	1.08 ($\pm$ 0.32)	10.69 ($\pm$ 1.48)	10.69 ($\pm$ 0.62)
$\mathbf{A}_3(t)$	2000	0.0	4.63 ($\pm$ 0.11)	4.66 ($\pm$ 0.11)	1.54 ($\pm$ 0.71)	1.52 ($\pm$ 0.71)	23.47 ($\pm$ 2.06)	1.91 ($\pm$ 1.56)
		0.3	4.63 ($\pm$ 0.11)	4.59 ($\pm$ 0.10)	1.75 ($\pm$ 0.81)	1.73 ($\pm$ 0.80)	23.53 ($\pm$ 2.03)	9.63 ($\pm$ 2.37)
		0.6	4.69 ($\pm$ 0.13)	4.32 ($\pm$ 0.11)	1.93 ($\pm$ 0.87)	1.86 ($\pm$ 0.88)	32.34 ($\pm$ 2.26)	18.91 ($\pm$ 1.69)
		0.9	4.69 ($\pm$ 0.14)	3.70 ($\pm$ 0.10)	2.07 ($\pm$ 0.91)	1.95 ($\pm$ 0.91)	32.29 ($\pm$ 2.68)	28.85 ($\pm$ 1.03)
	5000	0.0	2.72 ($\pm$ 0.05)	2.72 ($\pm$ 0.05)	0.94 ($\pm$ 0.33)	0.92 ($\pm$ 0.31)	14.38 ($\pm$ 1.22)	1.07 ($\pm$ 0.70)
		0.3	2.70 ($\pm$ 0.06)	2.63 ($\pm$ 0.05)	0.95 ($\pm$ 0.36)	0.94 ($\pm$ 0.34)	14.32 ($\pm$ 1.21)	5.05 ($\pm$ 1.21)
		0.6	2.70 ($\pm$ 0.06)	2.38 ($\pm$ 0.05)	1.05 ($\pm$ 0.38)	1.02 ($\pm$ 0.36)	14.40 ($\pm$ 1.34)	9.74 ($\pm$ 0.98)
		0.9	2.80 ($\pm$ 0.06)	1.69 ($\pm$ 0.04)	1.16 ($\pm$ 0.40)	1.09 ($\pm$ 0.35)	20.05 ($\pm$ 1.52)	14.99 ($\pm$ 0.61)
$\mathbf{A}_4(t)$	2000	0.0	3.33 ($\pm$ 0.09)	3.33 ($\pm$ 0.09)	1.48 ($\pm$ 0.70)	1.49 ($\pm$ 0.69)	26.11 ($\pm$ 2.14)	1.81 ($\pm$ 1.44)
		0.3	3.32 ($\pm$ 0.09)	3.23 ($\pm$ 0.08)	1.69 ($\pm$ 0.71)	1.65 ($\pm$ 0.71)	25.99 ($\pm$ 2.02)	5.93 ($\pm$ 2.42)
		0.6	3.32 ($\pm$ 0.09)	2.89 ($\pm$ 0.08)	1.86 ($\pm$ 0.77)	1.79 ($\pm$ 0.78)	25.97 ($\pm$ 2.46)	12.28 ($\pm$ 1.82)
		0.9	3.33 ($\pm$ 0.11)	2.06 ($\pm$ 0.07)	2.01 ($\pm$ 0.82)	1.91 ($\pm$ 0.80)	25.98 ($\pm$ 2.93)	18.68 ($\pm$ 1.01)
	5000	0.0	1.93 ($\pm$ 0.04)	1.93 ($\pm$ 0.04)	0.92 ($\pm$ 0.34)	0.89 ($\pm$ 0.33)	10.46 ($\pm$ 1.17)	1.09 ($\pm$ 0.70)
		0.3	1.90 ($\pm$ 0.05)	1.83 ($\pm$ 0.05)	0.93 ($\pm$ 0.31)	0.92 ($\pm$ 0.29)	10.47 ($\pm$ 1.14)	3.03 ($\pm$ 1.25)
		0.6	1.90 ($\pm$ 0.06)	1.60 ($\pm$ 0.05)	1.03 ($\pm$ 0.34)	1.01 ($\pm$ 0.31)	10.59 ($\pm$ 1.30)	5.68 ($\pm$ 0.96)
		0.9	1.90 ($\pm$ 0.06)	1.04 ($\pm$ 0.03)	1.13 ($\pm$ 0.37)	1.09 ($\pm$ 0.33)	10.72 ($\pm$ 1.54)	8.25 ($\pm$ 0.51)
$\mathbf{A}_5(t)$	2000	0.0	5.45 ($\pm$ 0.13)	5.49 ($\pm$ 0.13)	1.68 ($\pm$ 0.66)	1.67 ($\pm$ 0.70)	26.59 ($\pm$ 1.30)	2.02 ($\pm$ 1.32)
		0.3	5.47 ($\pm$ 0.16)	5.44 ($\pm$ 0.16)	1.95 ($\pm$ 0.88)	1.86 ($\pm$ 0.92)	26.57 ($\pm$ 1.45)	12.05 ($\pm$ 2.44)
		0.6	5.47 ($\pm$ 0.17)	5.20 ($\pm$ 0.16)	2.09 ($\pm$ 0.94)	1.98 ($\pm$ 1.00)	26.61 ($\pm$ 1.65)	24.82 ($\pm$ 2.24)
		0.9	5.48 ($\pm$ 0.18)	4.74 ($\pm$ 0.17)	2.22 ($\pm$ 0.97)	2.09 ($\pm$ 1.04)	26.69 ($\pm$ 1.91)	37.63 ($\pm$ 2.03)
	5000	0.0	3.16 ($\pm$ 0.05)	3.16 ($\pm$ 0.05)	0.95 ($\pm$ 0.43)	0.94 ($\pm$ 0.41)	16.81 ($\pm$ 1.06)	1.23 ($\pm$ 0.66)
		0.3	3.14 ($\pm$ 0.05)	3.06 ($\pm$ 0.05)	0.99 ($\pm$ 0.31)	0.98 ($\pm$ 0.28)	16.77 ($\pm$ 1.09)	6.14 ($\pm$ 1.25)
		0.6	3.13 ($\pm$ 0.06)	2.80 ($\pm$ 0.04)	1.09 ($\pm$ 0.34)	1.06 ($\pm$ 0.30)	16.83 ($\pm$ 1.21)	12.18 ($\pm$ 1.05)
		0.9	3.13 ($\pm$ 0.06)	2.18 ($\pm$ 0.03)	1.18 ($\pm$ 0.36)	1.13 ($\pm$ 0.32)	16.90 ($\pm$ 1.38)	18.17 ($\pm$ 0.74)
$\mathbf{A}_6(t)$	2000	0.0	4.46 ($\pm$ 0.10)	4.48 ($\pm$ 0.11)	1.58 ($\pm$ 0.78)	1.59 ($\pm$ 0.76)	25.20 ($\pm$ 1.96)	1.81 ($\pm$ 1.11)
		0.3	4.46 ($\pm$ 0.11)	4.41 ($\pm$ 0.12)	1.76 ($\pm$ 0.73)	1.72 ($\pm$ 0.73)	25.20 ($\pm$ 2.05)	8.76 ($\pm$ 2.61)
		0.6	4.46 ($\pm$ 0.11)	4.14 ($\pm$ 0.11)	1.92 ($\pm$ 0.79)	1.85 ($\pm$ 0.80)	25.16 ($\pm$ 2.46)	18.40 ($\pm$ 2.15)
		0.9	4.60 ($\pm$ 0.13)	3.57 ($\pm$ 0.12)	2.05 ($\pm$ 0.88)	1.94 ($\pm$ 0.87)	32.94 ($\pm$ 2.77)	27.95 ($\pm$ 1.33)
	5000	0.0	2.68 ($\pm$ 0.05)	2.68 ($\pm$ 0.05)	0.93 ($\pm$ 0.36)	0.90 ($\pm$ 0.34)	14.65 ($\pm$ 1.22)	1.15 ($\pm$ 0.76)
		0.3	2.66 ($\pm$ 0.05)	2.59 ($\pm$ 0.05)	0.96 ($\pm$ 0.32)	0.94 ($\pm$ 0.31)	14.62 ($\pm$ 1.25)	4.79 ($\pm$ 1.21)
		0.6	2.65 ($\pm$ 0.05)	2.33 ($\pm$ 0.05)	1.06 ($\pm$ 0.35)	1.02 ($\pm$ 0.32)	14.69 ($\pm$ 1.37)	9.41 ($\pm$ 0.98)
		0.9	2.79 ($\pm$ 0.06)	1.67 ($\pm$ 0.04)	1.17 ($\pm$ 0.38)	1.09 ($\pm$ 0.34)	20.11 ($\pm$ 1.53)	14.86 ($\pm$ 0.55)

Table 8: MAE errors ($\times 10^{-2}$) for Case 2(d) - $\sigma_3^2(t) = 0.5 + 0.5 \sin^2(\pi t)$.

function	n	ρ	MIAE		MAE		MIAE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
$\mathbf{A}_1(t)$	2000	0.0	3.03 ($\pm$ 0.09)	3.04 ($\pm$ 0.09)	1.38 ($\pm$ 0.59)	1.29 ($\pm$ 0.55)	15.01 ($\pm$ 1.82)	2.13 ($\pm$ 1.48)
		0.3	3.02 ($\pm$ 0.08)	2.96 ($\pm$ 0.08)	1.48 ($\pm$ 0.61)	1.41 ($\pm$ 0.60)	14.82 ($\pm$ 1.76)	6.41 ($\pm$ 2.66)
		0.6	3.03 ($\pm$ 0.09)	2.72 ($\pm$ 0.08)	1.61 ($\pm$ 0.68)	1.53 ($\pm$ 0.64)	14.76 ($\pm$ 2.15)	13.17 ($\pm$ 2.18)
		0.9	3.03 ($\pm$ 0.09)	2.19 ($\pm$ 0.07)	1.74 ($\pm$ 0.74)	1.64 ($\pm$ 0.65)	14.73 ($\pm$ 2.57)	20.00 ($\pm$ 1.39)
	5000	0.0	2.03 ($\pm$ 0.05)	2.04 ($\pm$ 0.05)	0.78 ($\pm$ 0.30)	0.77 ($\pm$ 0.32)	3.64 ($\pm$ 0.97)	1.21 ($\pm$ 0.88)
		0.3	2.00 ($\pm$ 0.05)	1.98 ($\pm$ 0.05)	0.82 ($\pm$ 0.26)	0.79 ($\pm$ 0.25)	3.57 ($\pm$ 0.92)	3.76 ($\pm$ 1.33)
		0.6	2.00 ($\pm$ 0.05)	1.84 ($\pm$ 0.05)	0.91 ($\pm$ 0.29)	0.86 ($\pm$ 0.27)	3.67 ($\pm$ 0.98)	7.37 ($\pm$ 1.07)
		0.9	2.00 ($\pm$ 0.05)	1.57 ($\pm$ 0.03)	0.99 ($\pm$ 0.31)	0.93 ($\pm$ 0.29)	3.81 ($\pm$ 1.07)	10.93 ($\pm$ 0.66)
$\mathbf{A}_2(t)$	2000	0.0	2.98 ($\pm$ 0.09)	3.00 ($\pm$ 0.09)	1.33 ($\pm$ 0.62)	1.24 ($\pm$ 0.61)	15.17 ($\pm$ 1.84)	1.86 ($\pm$ 1.54)
		0.3	2.98 ($\pm$ 0.08)	2.92 ($\pm$ 0.09)	1.50 ($\pm$ 0.59)	1.43 ($\pm$ 0.59)	15.01 ($\pm$ 1.79)	6.22 ($\pm$ 2.33)
		0.6	2.98 ($\pm$ 0.09)	2.68 ($\pm$ 0.08)	1.64 ($\pm$ 0.64)	1.54 ($\pm$ 0.65)	14.96 ($\pm$ 2.16)	12.81 ($\pm$ 2.16)
		0.9	2.98 ($\pm$ 0.09)	2.11 ($\pm$ 0.07)	1.78 ($\pm$ 0.70)	1.65 ($\pm$ 0.66)	14.92 ($\pm$ 2.56)	19.60 ($\pm$ 1.53)
	5000	0.0	1.98 ($\pm$ 0.04)	1.99 ($\pm$ 0.04)	0.78 ($\pm$ 0.31)	0.77 ($\pm$ 0.33)	4.30 ($\pm$ 0.79)	1.32 ($\pm$ 0.69)
		0.3	1.96 ($\pm$ 0.04)	1.93 ($\pm$ 0.04)	0.81 ($\pm$ 0.25)	0.76 ($\pm$ 0.26)	4.23 ($\pm$ 0.81)	3.70 ($\pm$ 1.30)
		0.6	1.95 ($\pm$ 0.04)	1.78 ($\pm$ 0.04)	0.90 ($\pm$ 0.27)	0.83 ($\pm$ 0.28)	4.30 ($\pm$ 0.88)	7.30 ($\pm$ 1.05)
		0.9	1.95 ($\pm$ 0.05)	1.44 ($\pm$ 0.03)	0.98 ($\pm$ 0.30)	0.89 ($\pm$ 0.29)	4.41 ($\pm$ 0.98)	10.86 ($\pm$ 0.69)
$\mathbf{A}_3(t)$	2000	0.0	4.00 ($\pm$ 0.10)	4.01 ($\pm$ 0.10)	1.37 ($\pm$ 0.61)	1.34 ($\pm$ 0.58)	21.36 ($\pm$ 1.46)	2.23 ($\pm$ 1.52)
		0.3	4.00 ($\pm$ 0.09)	3.94 ($\pm$ 0.09)	1.55 ($\pm$ 0.65)	1.50 ($\pm$ 0.71)	21.30 ($\pm$ 1.65)	9.07 ($\pm$ 2.43)
		0.6	4.00 ($\pm$ 0.10)	3.70 ($\pm$ 0.08)	1.68 ($\pm$ 0.71)	1.60 ($\pm$ 0.78)	21.23 ($\pm$ 1.96)	18.91 ($\pm$ 1.81)
		0.9	4.00 ($\pm$ 0.11)	3.21 ($\pm$ 0.08)	1.79 ($\pm$ 0.77)	1.67 ($\pm$ 0.80)	21.17 ($\pm$ 2.27)	28.79 ($\pm$ 1.11)
	5000	0.0	2.38 ($\pm$ 0.05)	2.38 ($\pm$ 0.05)	0.80 ($\pm$ 0.25)	0.79 ($\pm$ 0.29)	9.03 ($\pm$ 0.96)	1.14 ($\pm$ 0.74)
		0.3	2.37 ($\pm$ 0.05)	2.31 ($\pm$ 0.05)	0.83 ($\pm$ 0.29)	0.79 ($\pm$ 0.31)	8.95 ($\pm$ 0.96)	5.00 ($\pm$ 1.20)
		0.6	2.40 ($\pm$ 0.05)	2.08 ($\pm$ 0.04)	0.92 ($\pm$ 0.31)	0.86 ($\pm$ 0.32)	13.69 ($\pm$ 1.03)	9.88 ($\pm$ 0.93)
		0.9	2.40 ($\pm$ 0.05)	1.48 ($\pm$ 0.03)	1.00 ($\pm$ 0.32)	0.91 ($\pm$ 0.31)	13.71 ($\pm$ 1.14)	14.73 ($\pm$ 0.59)
$\mathbf{A}_4(t)$	2000	0.0	2.75 ($\pm$ 0.08)	2.75 ($\pm$ 0.08)	1.34 ($\pm$ 0.60)	1.24 ($\pm$ 0.56)	17.41 ($\pm$ 1.85)	1.98 ($\pm$ 1.59)
		0.3	2.74 ($\pm$ 0.07)	2.66 ($\pm$ 0.07)	1.46 ($\pm$ 0.61)	1.40 ($\pm$ 0.62)	17.24 ($\pm$ 1.81)	5.47 ($\pm$ 2.38)
		0.6	2.75 ($\pm$ 0.08)	2.38 ($\pm$ 0.06)	1.60 ($\pm$ 0.68)	1.52 ($\pm$ 0.66)	17.19 ($\pm$ 2.18)	11.14 ($\pm$ 2.00)
		0.9	2.75 ($\pm$ 0.09)	1.69 ($\pm$ 0.05)	1.74 ($\pm$ 0.73)	1.62 ($\pm$ 0.67)	17.15 ($\pm$ 2.60)	16.99 ($\pm$ 1.07)
	5000	0.0	1.62 ($\pm$ 0.04)	1.62 ($\pm$ 0.04)	0.78 ($\pm$ 0.28)	0.77 ($\pm$ 0.30)	6.40 ($\pm$ 1.03)	1.20 ($\pm$ 0.78)
		0.3	1.60 ($\pm$ 0.05)	1.55 ($\pm$ 0.04)	0.81 ($\pm$ 0.25)	0.77 ($\pm$ 0.26)	6.33 ($\pm$ 0.98)	2.74 ($\pm$ 1.19)
		0.6	1.60 ($\pm$ 0.05)	1.36 ($\pm$ 0.04)	0.90 ($\pm$ 0.28)	0.85 ($\pm$ 0.28)	6.39 ($\pm$ 1.06)	5.22 ($\pm$ 0.96)
		0.9	1.60 ($\pm$ 0.05)	0.91 ($\pm$ 0.03)	0.98 ($\pm$ 0.31)	0.92 ($\pm$ 0.29)	6.46 ($\pm$ 1.21)	7.66 ($\pm$ 0.51)
$\mathbf{A}_5(t)$	2000	0.0	4.80 ($\pm$ 0.11)	4.84 ($\pm$ 0.11)	1.49 ($\pm$ 0.58)	1.44 ($\pm$ 0.58)	15.36 ($\pm$ 1.29)	2.16 ($\pm$ 1.55)
		0.3	4.81 ($\pm$ 0.13)	4.79 ($\pm$ 0.13)	1.71 ($\pm$ 0.73)	1.59 ($\pm$ 0.81)	15.14 ($\pm$ 1.36)	12.72 ($\pm$ 2.63)
		0.6	4.81 ($\pm$ 0.14)	4.61 ($\pm$ 0.14)	1.83 ($\pm$ 0.77)	1.71 ($\pm$ 0.87)	15.17 ($\pm$ 1.47)	26.05 ($\pm$ 2.49)
		0.9	4.81 ($\pm$ 0.15)	4.26 ($\pm$ 0.15)	1.93 ($\pm$ 0.81)	1.80 ($\pm$ 0.91)	15.22 ($\pm$ 1.61)	39.43 ($\pm$ 2.29)
	5000	0.0	2.77 ($\pm$ 0.05)	2.77 ($\pm$ 0.05)	0.82 ($\pm$ 0.33)	0.80 ($\pm$ 0.38)	10.16 ($\pm$ 0.87)	1.34 ($\pm$ 0.74)
		0.3	2.75 ($\pm$ 0.05)	2.69 ($\pm$ 0.04)	0.87 ($\pm$ 0.24)	0.83 ($\pm$ 0.26)	10.10 ($\pm$ 0.96)	6.26 ($\pm$ 1.32)
		0.6	2.74 ($\pm$ 0.05)	2.48 ($\pm$ 0.04)	0.96 ($\pm$ 0.27)	0.90 ($\pm$ 0.27)	10.12 ($\pm$ 1.02)	12.57 ($\pm$ 1.10)
		0.9	2.74 ($\pm$ 0.05)	1.99 ($\pm$ 0.03)	1.04 ($\pm$ 0.29)	0.96 ($\pm$ 0.28)	10.15 ($\pm$ 1.13)	18.86 ($\pm$ 0.77)
$\mathbf{A}_6(t)$	2000	0.0	3.93 ($\pm$ 0.09)	3.94 ($\pm$ 0.09)	1.41 ($\pm$ 0.67)	1.40 ($\pm$ 0.68)	15.22 ($\pm$ 1.60)	1.94 ($\pm$ 1.20)
		0.3	3.93 ($\pm$ 0.10)	3.88 ($\pm$ 0.10)	1.54 ($\pm$ 0.63)	1.48 ($\pm$ 0.61)	15.11 ($\pm$ 1.83)	8.87 ($\pm$ 2.71)
		0.6	3.96 ($\pm$ 0.10)	3.65 ($\pm$ 0.10)	1.65 ($\pm$ 0.72)	1.57 ($\pm$ 0.70)	21.89 ($\pm$ 2.07)	18.05 ($\pm$ 2.10)
		0.9	3.95 ($\pm$ 0.11)	3.11 ($\pm$ 0.09)	1.77 ($\pm$ 0.77)	1.65 ($\pm$ 0.73)	21.83 ($\pm$ 2.36)	27.72 ($\pm$ 1.50)
	5000	0.0	2.37 ($\pm$ 0.04)	2.36 ($\pm$ 0.04)	0.79 ($\pm$ 0.29)	0.79 ($\pm$ 0.30)	9.31 ($\pm$ 0.96)	1.27 ($\pm$ 0.78)
		0.3	2.35 ($\pm$ 0.05)	2.29 ($\pm$ 0.04)	0.84 ($\pm$ 0.26)	0.80 ($\pm$ 0.28)	9.26 ($\pm$ 0.96)	4.70 ($\pm$ 1.24)
		0.6	2.35 ($\pm$ 0.05)	2.09 ($\pm$ 0.04)	0.93 ($\pm$ 0.28)	0.86 ($\pm$ 0.29)	9.29 ($\pm$ 1.02)	9.40 ($\pm$ 1.02)
		0.9	2.40 ($\pm$ 0.05)	1.49 ($\pm$ 0.03)	1.01 ($\pm$ 0.30)	0.92 ($\pm$ 0.30)	13.77 ($\pm$ 1.15)	14.57 ($\pm$ 0.58)

EMPIRICAL EXPERIMENTS

Dataset Information. The glaucoma real-world appraisal progression ensemble (GRAPE) is a longitudinal multimodal dataset collected for glaucoma management. It contains fundus images together with clinical information collected during follow-up. For each image visit, two image types data are collected: a color fundus photograph (CFP), an optic-disc-centered region-of-interest (ROI) image. An example is given in Figure 5.

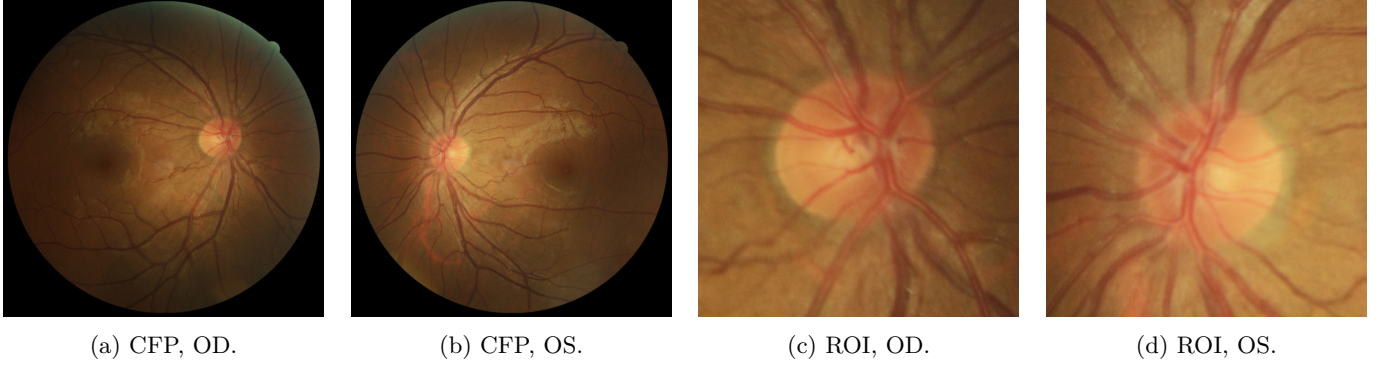


Figure 5: Original CFP and ROI images from an representative paired visit.

The intraocular pressure (IOP), age, gender, and visual-field (VF) measurements are also included in the dataset. Since elevated IOP is an important biomarker and modifiable risk factor for glaucoma progression, our objective is to quantify how the association between fundus images and IOP varies with age and to evaluate the predictive value of the image and clinical covariates.

Preprocessing. The distributed image subset contains 631 eye-level visits with matched CFP and ROI images. We paired oculus dexter (OD) and oculus sinister (OS) records from the same subject at the same follow-up time, obtaining 276 paired visits. After excluding a paired visit whenever the IOP of either eye exceeded 35 mmHg, the primary analysis retained 273 paired visits, corresponding to 546 eye-level observations for each image representation. All images were resized to $192 \times 192 \times 3$, and OS images were horizontally flipped to align with the OD images.

Let $i = 1, \dots, 273$ index paired visits and $j \in \{1, 2\}$ index the OD and OS eyes, respectively. For either the CFP or ROI analysis, let $\mathcal{X}_{ij} \in \mathbb{R}^{192 \times 192 \times 3}$ denote the image tensor and let y_{ij} denote the corresponding IOP. The index variable t_i is age, min-max normalized to $[0, 1]$. The pair-level covariate vector $\mathbf{z}_i \in \mathbb{R}^{60}$ contains gender and the paired-eye averages of 59 non-blind-spot VF measurements; the continuous VF covariates and IOP outcomes are standardized before model fitting. We then apply the paired-eye VCTR model

$$y_{ij} = \langle \mathcal{X}_{ij}, \mathcal{A}(t_i) \rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij}, \quad \text{where} \quad \begin{bmatrix} \epsilon_{i1} \\ \epsilon_{i2} \end{bmatrix} \sim \mathbf{N} \left(\mathbf{0}, \sigma^2(t_i) \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix} \right),$$

and $\mathcal{A}(t)$ captures the age-dependent image effect and $\boldsymbol{\beta}$ contains the constant effects of the clinical covariates.

Choosing Hyperparameter. We select four hyperparameters separately for CFP and ROI: the partition grid $S^{(1)} \times S^{(2)} \times S^{(3)}$, the blockwise CP rank R , the bandwidth h for estimating the varying image effect, and the bandwidth $\bar{h}$ for estimating the variance function. The candidate feature set contains 17 partition grids and 56 available (S, R) combinations. For each feature configuration, we consider 32 values of h ranging from 0.18 to 8.00 and 15 values of $\bar{h}$ ranging from 0.20 to 4.00. This gives $56 \times 32 \times 15 = 26,880$ candidates for each image type and 53,760 candidates in total. Hyperparameters are selected using 5-fold cross-validation.

Table 9 reports, for each (S, R) configuration, the smallest cross-validated RMSE over all 480 combinations of $(h, \bar{h})$. The selected configurations are $S^{(1)} \times S^{(2)} \times S^{(3)} = 3 \times 4 \times 1$, $R = 1$, $h = 1.80$, and $\bar{h} = 0.25$ for CFP, and $6 \times 2 \times 1$, $R = 1$, $h = 0.85$, and $\bar{h} = 0.30$ for ROI. Their standardized-scale RMSE values are 0.9890 and 0.9620.

Table 9: Minimum RMSE by tensor partition and CP rank.

$S^{(1)} \times S^{(2)} \times S^{(3)}$	CFP				ROI			
	$R = 1$	$R = 2$	$R = 3$	$R = 4$	$R = 1$	$R = 2$	$R = 3$	$R = 4$
$2 \times 2 \times 1$	0.9958	0.9993	1.0068	1.0125	0.9793	0.9746	0.9942	0.9964
$2 \times 3 \times 1$	1.0032	1.0213	1.0207	–	0.9849	0.9942	1.0219	–
$2 \times 4 \times 1$	1.0096	1.0219	1.0212	–	0.9924	1.0089	1.0370	–
$2 \times 6 \times 1$	1.0204	1.0369	1.1059	–	1.0093	1.0410	1.0448	–
$3 \times 2 \times 1$	0.9931	0.9993	1.0247	–	0.9742	0.9936	1.0190	–
$3 \times 3 \times 1$	0.9964	1.0261	1.0383	1.0440	0.9782	0.9978	1.0563	1.0698
$3 \times 4 \times 1$	0.9890	1.0025	1.0547	–	1.0054	1.0521	1.0905	–
$3 \times 6 \times 1$	1.0087	1.0596	1.1367	–	1.0112	1.0890	1.1446	–
$4 \times 2 \times 1$	1.0055	1.0135	1.0326	–	0.9766	0.9885	1.0242	–
$4 \times 3 \times 1$	1.0054	1.0626	1.1094	–	0.9823	1.0246	1.0476	–
$4 \times 4 \times 1$	1.0131	1.0382	1.1292	1.1915	1.0023	1.0397	1.1207	1.1167
$4 \times 6 \times 1$	1.0363	1.1582	1.2898	–	0.9768	1.0598	1.1639	–
$6 \times 2 \times 1$	1.0160	1.0358	1.0586	–	0.9620	0.9792	1.0512	–
$6 \times 3 \times 1$	1.0231	1.0709	1.1856	–	0.9904	1.0589	1.1264	–
$6 \times 4 \times 1$	1.0231	1.0666	1.1431	–	1.0174	1.1024	1.2372	–
$6 \times 6 \times 1$	1.0701	1.3510	1.7346	2.2838	1.0444	1.1539	1.3007	1.7721
$8 \times 8 \times 1$	1.1220	1.7289	3.4191	5.0691	1.1446	1.4350	3.5168	4.7495

Figure 6 shows an image example under our chosen hyperparameter.

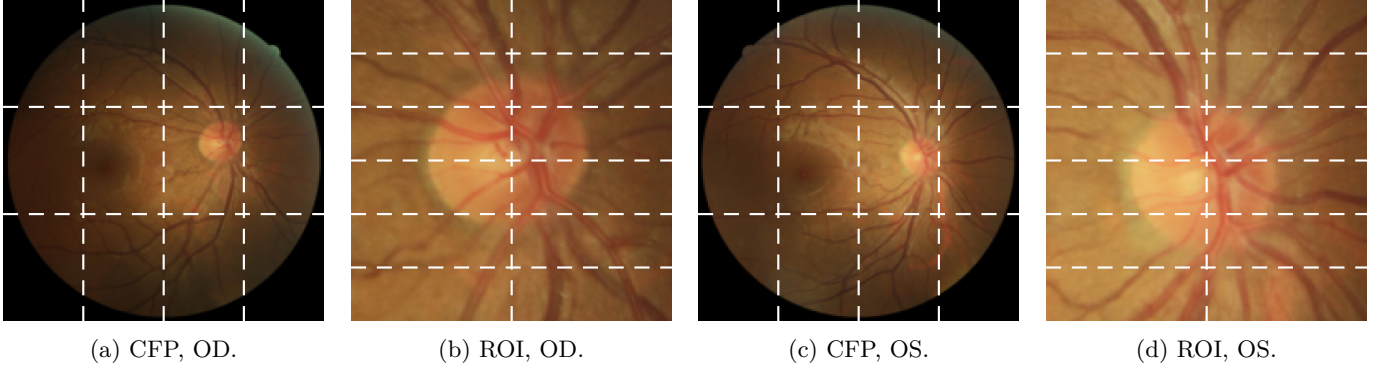


Figure 6: Representative paired-eye GRAPE images with partition $3 \times 4 \times 1$ for CFP and $6 \times 2 \times 1$ for ROI.

Model Interpretation Figure 7 presents the estimated coefficient functions $\hat{a}_{1s}(t)$ from the ROI VCTR model. The model uses the prediction-selected ROI image configuration $S^{(1)} \times S^{(2)} \times S^{(3)} = 6 \times 2 \times 1$, $R = 1$, $h = 0.85$, and $\bar{h} = 0.30$. The confidence bands are omitted from this figure to make the estimated age-dependent trajectories easier to compare across the 12 image blocks.

ROI X+Z paired VCTR: coefficient functions

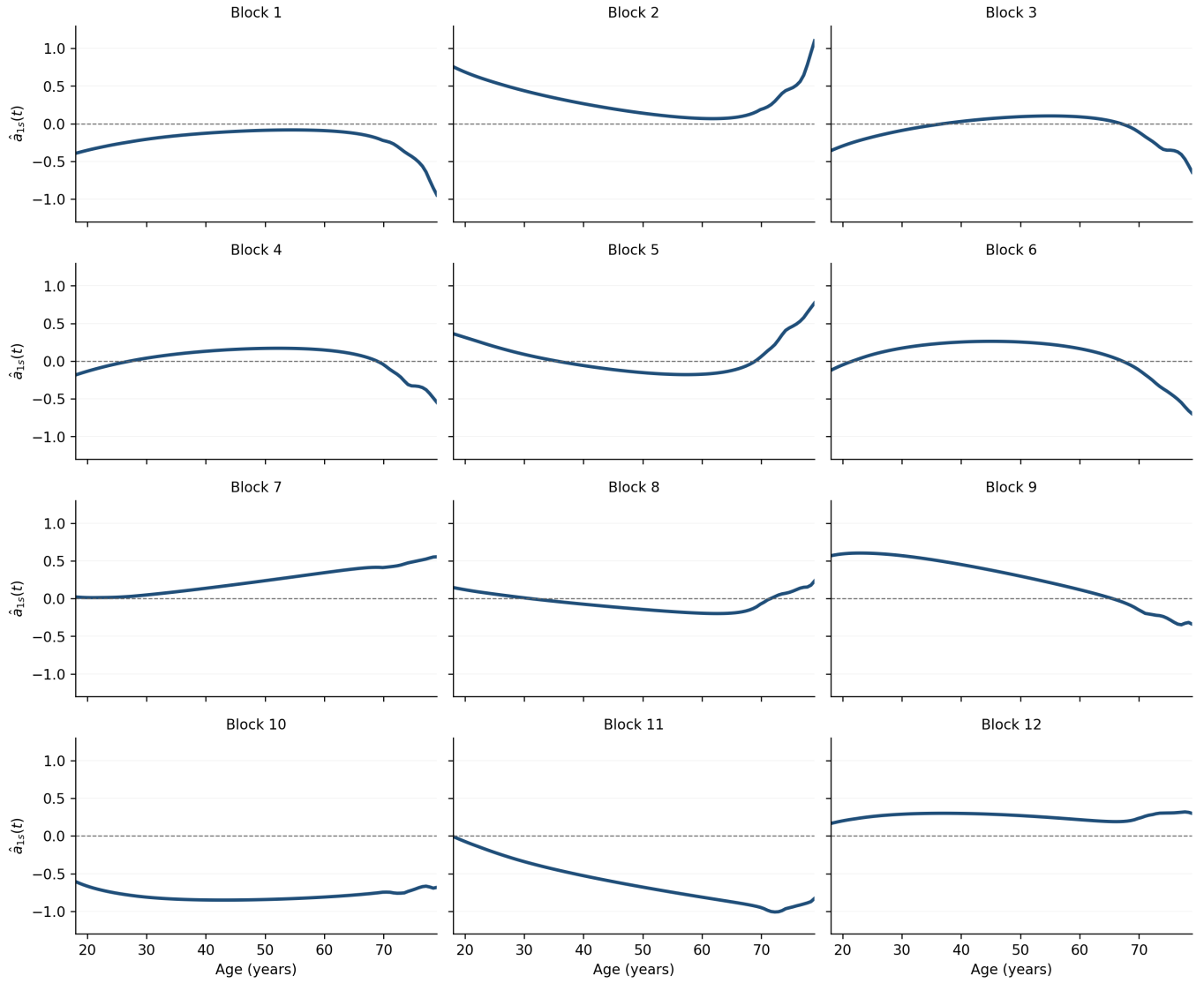


Figure 7: Estimated ROI coefficient functions from the VCTR model.

Table 10: All scalar coefficients in ROI VCTR model based on a $B = 2000$ bootstrap.

Covariate	Estimate	Bootstrap SE	95% CI	p -value	Covariate	Estimate	Bootstrap SE	95% CI	p -value
Gender	0.3636	0.7484	[-1.0519, 1.8401]	0.543	VF30	-1.5992	1.8574	[-5.5043, 1.6875]	0.294
VF00	-4.7368	1.8050	[-7.7807, -0.7137]	0.023	VF31	0.7623	2.0298	[-2.7546, 5.1378]	0.606
VF01	2.4040	1.6152	[-1.0620, 5.2806]	0.148	VF33	-1.4154	1.2216	[-3.9614, 0.6820]	0.190
VF02	1.2436	2.0067	[-2.9930, 4.9205]	0.522	VF34	0.9300	1.1479	[-1.4252, 3.1472]	0.466
VF03	3.8123	2.2362	[-0.3617, 8.3195]	0.070	VF35	-2.4346	1.5201	[-5.1943, 0.7145]	0.144
VF04	4.6147	2.0020	[0.7861, 8.5312]	0.019	VF36	0.7484	1.4691	[-2.2744, 3.6347]	0.676
VF05	-3.2855	2.2925	[-8.2329, 1.0029]	0.112	VF37	-1.1999	2.1288	[-5.5216, 2.9894]	0.556
VF06	-1.2673	2.3895	[-6.5481, 2.9152]	0.447	VF38	1.5452	1.8560	[-2.1044, 5.1512]	0.386
VF07	1.9111	2.8446	[-3.6889, 7.6151]	0.570	VF39	-2.2188	1.7798	[-5.7668, 1.3339]	0.224
VF08	-2.1027	2.1817	[-6.6136, 1.7527]	0.295	VF40	0.8376	1.5046	[-1.9421, 3.9402]	0.549
VF09	0.6285	2.1515	[-3.4967, 5.0914]	0.689	VF41	-0.9182	1.6364	[-4.0978, 2.4663]	0.556
VF10	-0.1016	1.7733	[-3.7164, 3.2306]	0.906	VF42	0.6432	2.4218	[-3.5035, 6.0623]	0.706
VF11	0.9169	1.7497	[-2.4376, 4.6234]	0.502	VF43	0.7816	2.1333	[-2.8436, 5.5297]	0.600
VF12	-0.7978	2.2431	[-4.4937, 4.3516]	0.886	VF44	-2.1332	1.9133	[-5.2422, 2.0390]	0.341
VF13	3.9601	2.2563	[-0.8856, 7.9618]	0.119	VF45	1.2080	1.5056	[-1.4518, 4.4074]	0.352
VF14	0.4009	2.0320	[-1.8102, 6.2806]	0.758	VF46	-2.2190	1.8511	[-5.7743, 1.3990]	0.222
VF15	-1.7483	2.6266	[-7.2105, 3.0866]	0.454	VF47	2.2628	2.2334	[-1.8410, 7.0308]	0.241
VF16	4.7956	2.7778	[-0.0749, 10.6414]	0.053	VF48	1.2110	2.1274	[-2.6759, 5.6408]	0.504
VF17	-3.2574	2.9900	[-9.2973, 2.1048]	0.286	VF49	-1.9245	1.7834	[-5.6493, 1.4426]	0.227
VF18	-2.7720	2.3446	[-7.2509, 2.0119]	0.253	VF50	-0.2721	1.2640	[-2.5242, 2.3640]	0.928
VF19	2.8077	2.2909	[-1.8514, 6.9746]	0.252	VF51	-0.4740	1.0120	[-2.6796, 1.2211]	0.508
VF20	-0.5433	2.3672	[-5.0245, 4.2213]	0.826	VF52	-1.3907	1.1003	[-3.7335, 0.5389]	0.167
VF22	1.8373	1.8401	[-1.7735, 5.5462]	0.298	VF53	0.5845	1.5011	[-2.2244, 3.7814]	0.629
VF23	-0.7241	1.5016	[-3.6695, 2.1683]	0.624	VF54	1.6629	1.9435	[-2.1976, 5.2682]	0.457
VF24	-2.4735	2.0545	[-6.5204, 1.3948]	0.229	VF55	0.8503	2.2200	[-3.6169, 5.1068]	0.795
VF25	-0.0420	2.6614	[-5.2189, 5.5348]	0.986	VF56	0.4785	2.1697	[-4.4390, 4.2848]	0.926
VF26	-1.9174	2.0246	[-5.9303, 2.3030]	0.345	VF57	1.9185	1.8760	[-2.3761, 4.8878]	0.422
VF27	-4.6170	2.7248	[-9.7895, 1.1205]	0.096	VF58	0.8606	1.8375	[-2.9659, 4.2252]	0.752
VF28	4.5282	2.9070	[-2.3001, 8.8359]	0.202	VF59	1.1116	1.8124	[-2.2541, 4.8736]	0.488
VF29	-4.6475	2.6019	[-9.7920, 0.3106]	0.067	VF60	2.4809	1.3808	[0.0853, 5.3715]	0.042

Figure 8 shows the full-sample estimate of the age-varying residual standard deviation used in the paired covariance model with variance bandwidth $\bar{h} = 0.30$.

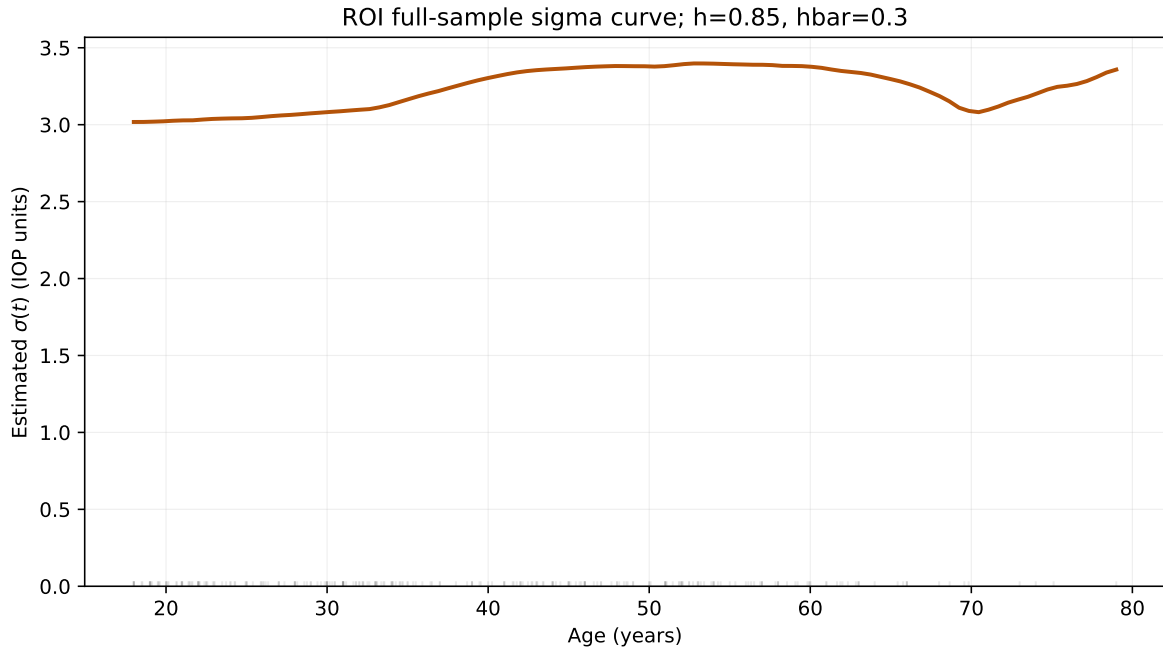


Figure 8: Full-sample estimate of $\sigma(t)$ for the ROI paired VCTR model.